



Al-Raba'a, Basem Ibrahim Malawi & Kitagawa, Yoshihisa. 2022. Reflexivity, reciprocity and collectivity in Jordanian Arabic. *Glossa: a journal of general linguistics* 7(1). pp. 1–38. DOI: <https://doi.org/10.16995/glossa.5799>



Open Library of Humanities

Reflexivity, reciprocity and collectivity in Jordanian Arabic

Basem Ibrahim Malawi Al-Raba'a, KIMEP University, Kazakhstan, b_al-rabaa@ju.edu.jo

Yoshihisa Kitagawa, Indiana University Bloomington, USA, kitagawa@iu.edu

This study examines, first, the syntactic asymmetries between reflexive and reciprocal constructions in Jordanian Arabic (JA), paying special attention to those with morphological marking. In particular, it will be argued that morphologically marked reflexives are one-place predicates, but their reciprocal counterparts are two-place predicates, with their internal arguments being realized either overtly or covertly in syntax. It will then be argued that what is generally regarded as a reciprocalizing morpheme (t-) indicates obligatory “collectivity,” namely, two subeventualities significantly overlapping in time (and place). An attempt will also be made to identify the semantic apparatuses which can properly capture the interpretation of a sentence when a T-marked verb appears. Finally, it will also be inquired how the T-morpheme derives the so-called Verb Forms 5, 6 and 8, and exhibits either reflexive or collective interpretations in JA. A brief cross-language and cross-dialect survey will also be attempted in Appendix B and Appendix C.

Appendices are placed after "Acknowledgments/Competing interests" on p. 35.



1. Introduction — Reflexives and Reciprocals in JA

As is well known, reflexivity and reciprocity represent two different but related linguistic concepts, and we often witness more than one way of their grammaticalization. In Jordanian Arabic (JA), for example, reflexivization may be achieved in at least two ways. As in (1) below, it may be represented syntactically with the internal argument of the verb appearing as a SELF-anaphor.¹

- (1) l-walad₁ ?addab nafs-uh₁.
 the-boy.NOM behaved self-him.ACC
 ‘The boy behaved himself.’

As shown in (2) below, it may also be represented morphologically when a reflexive morpheme (e.g., *t-*) is affixed to the verb.

- (2) l-walad t-?addab.
 the-boy.NOM T-behaved
 ‘The boy behaved. (= The boy behaved himself.)’

Similarly, reciprocal constructions in JA fall into two types, one making use of reciprocal anaphors as in (3) and the other involving overt morphological marking on the verb as in (4).²

- (3) l-waladeen_{<1+2>} ?aawanu baʕḏ-hum l-baʕḏ_{<1↔2>}.
 the-boys.DL.NOM helped some-them.ACC the-some.ACC
 ‘The two boys helped each other.’
- (4) l-waladeen t-ʔaawanu.
 the-boys.DL.NOM T-helped
 ‘The two boys helped each other.’

In the remainder of this study, for the sake of convenience, we label the constructions as in (1) and (3) “syntactic reflexives/reciprocals” and those as in (2) and (4) “morphological reflexives/reciprocals,” paying attention to the distinct surface forms of the involved verbs. These contrastive labelings, however, will turn out to be too loose eventually as our analyses proceed.

¹ JA taken up in this work is what may be referred to as Rural Jordanian Arabic, which the first author speaks. The judgments provided on the JA examples, which were presented with relevant discourses and/or contexts, have also been agreed upon by at least four (sometimes five) other native speakers. The glossing conventions are as follows: ACC = accusative, DIST = distributive, DL = dual, F = feminine, GEN = genitive, M = masculine, NEG = negation, NOM = nominative, OM = object marker, RECIP = reciprocal, REFL = reflexive.

² The reciprocal interpretation of the anaphor is indicated in this work by coindexation accompanied by a bidirectional arrow as in _{<1↔2>}. It is meant to indicate the coindexation with the two-membered set _{<1+2>} and the crossing relation holding between the two members of this set. Reciprocity of course may hold among three or more members of the set, but for simplicity’s sake, we will discuss only the reciprocity holding between two members. We are concerned in this work, in other words, mostly with the so-called “strong reciprocity.” We will briefly discuss “weak reciprocity” in Appendix C.

In traditional Arabic grammar, reflexive constructions as in (1)–(2) and reciprocal constructions as in (3)–(4) are generally considered to involve a parallel structure. That is, reflexive/reciprocal verbs lacking overt morphological marking as in (1) and (3) are treated as transitive and must therefore take an anaphor as their object. In contrast, morphologically marked verbs as in (2) and (4) are analyzed as intransitive, reflexive or reciprocal (see Wright 1896; Holes 2004; Ryding 2005). Similar analyses are often adopted for other languages like Greek and Hebrew as well (e.g., Reinhart & Siloni 2005; Dimitriadis 2008b; Siloni 2008; 2012).

There is one issue that has caused controversy in the literature when this transitive-intransitive dichotomy is adopted. It is how the morphologically marked one-place construction can be assimilated to the syntactically represented two-place construction in providing their interpretations. Reinhart & Siloni (2005), for instance, proposed that “morphological” reflexive constructions in Hebrew like (5b) below be analyzed as involving two syntactic operations which let this sentence be interpreted on a par with (5a): what they call “reflexivization bundling” and accusative Case absorption.

- (5) a. Dan raxac et **acmo**.
 Dan washed OM **himself**
 ‘Dan washed himself.’
 b. Dan **hitraxec**.
 Dan **REFL**.washed
 ‘Dan washed.’ (Reinhart & Siloni 2005: 390)

They hypothesize that the bundling operation first fuses the external theta-role of the two-place verb with its internal one (e.g., Agent and Theme) forming one complex theta-role, and then assigns this complex theta-role to the sole syntactic argument, the subject. As a result, the verb is realized as a one-place predicate, and the internal argument need not be syntactically projected or case-marked while it is expressing a two-place thematic relation.

Pushing the same idea, Siloni (2001; 2008; 2012) further contends that a reciprocal construction in Hebrew like (6b) below undergoes a “reciprocalization bundling” operation (by which the external and internal theta-roles are fused and assigned to the subject argument) as well as absorption of accusative Case (see also Laks 2007), and becomes interpreted on a par with (6a).

- (6) a. Hem ni]ku { **ze** et **ze** / **exad** et **ha-]eni** }.
 they kissed this ACC this one ACC the-second
 ‘They kissed each other.’
 b. Hem **hitna]ku**.
 they **RECIP**.kissed
 ‘They kissed.’ (Siloni 2008: 452)

Contrary to the studies described just above, LeTourneau (1998) argues that a morphologically marked reciprocal in Standard Arabic involves a phonetically null internal argument. To make a long story short, he proposes that the implicit reciprocal ($[e]_{\text{RECIP}}$) base-generated in the object position in (7a) below is syntactically raised to the left periphery of the verbal stem. The moved $[e]_{\text{RECIP}}$ in this position then is claimed to be morphologically realized as the prefix *ta-*, yielding the structure in (7b). (The data below from Standard Arabic are adapted from LeTourneau 1998: 106–109, 111.)

- (7) a. aṭ-ṭulaab-u qaabaluu [e]_{RECIP}.
 the-students-NOM met
 ↑
 ‘The students met each other.’
- b. aṭ-ṭulaab-u ta-qaabaluu.
 the-students-NOM TA-met
 ‘The students met each other.’

Bar-Asher Siegal (2016: 26–27) also offered a semantic motivation for postulating an implicit internal argument of the so-called reciprocal predicates in Hebrew, which also appear with a T-morpheme (in the form *hit-V*). First, comparing the interpretations of (8a) and (8b) below, he points out that reciprocal interpretations in this construction arise due to the pragmatic relation holding between the two arguments rather than the morphological function of *hit-V*. (Note the absence of reciprocity in (8b).)

- (8) a. Josi **hit**xabek im Rina.³
 Yosi *HIT*-hugged with Rina
 ‘Yosi and Rina hugged each other.’
- b. Josi **hit**xabek im ha-karit.
 Yosi *HIT*-hugged with the-pillow
 ‘Yosi hugged the pillow.’

He then characterizes *hit-V* as a dyadic predicate that obligatorily selects both external and internal arguments, as described in (9).

- (9) *hit-V*: (*hit-Ver*, *hit-Ved*) (N.B. *-er* as in *killer* and *-ed* as in *killed* in English)

This enforces the presence of a phonetically empty implicit internal argument even when an overt object is missing as in (10a) and (10c).

³ The reciprocal interpretation implied as in (8a) is referred to in the literature as “discontinuous reciprocal” (Dimitriadis 2008a; 2008b; Siloni 2008; Bar-Asher Siegal 2016). We will return to this linguistic concept later and discuss it in relation to Arabic.

- (10) a. **Josi ve-Rina hitnaʃku [e]_{RECIP}**
 Yosi and-Rina HIT-kissed
 ‘Yosi and Rina kissed each other.’
- b. **Josi hitnaʃek im Rina.**
 Yosi HIT-kissed with Rina
 ‘Yosi and Rina kissed each other.’
- c. **Rina hitnaʃqa [e] ha-jom b-a-paʔam ha-riʃona.**
 Rina HIT-kissed the-day in-the-time the-first
 ‘Rina and someone kissed each other today for the first time.’

Bar-Asher Siegal points out that this analysis permits us to capture all the interpretations in (8a–b) and (10a–c) uniformly, whether they involve a transitive or an intransitive predicate, a plural or a singular subject, or a reciprocal or a non-reciprocal interpretation. Note that the so-called oblique argument is regarded as serving as the internal argument of the predicate in (10b) and presumably in (10a) and (10c) as well, whether it is overt or covert. These cross-language studies encourage us to hypothesize that the optionality of the oblique reciprocal argument in (11) below in fact is analyzable in two ways as illustrated in (12), in which either an overt or empty reciprocal anaphor appears as the complement of a two-place verb in JA.^{4,5}

- (11) l-waladeen_{<1+2>} t-ʃaawanu (**maʃ baʃð-hum 1-baʃð**_{<1↔2>}).
 the-boys.DL.NOM T-helped with some-them.GEN the-some.GEN
 ‘The two boys helped each other.’
- (12) l-waladeen_{<1+2>} t-ʃaawanu {**maʃ baʃð-hum 1-baʃð**_{<1↔2>} / **[e]_{RECIP<1↔2>}**}.
 the-boys.DL.NOM T-helped with some-them.GEN the-some.GEN
 ‘The two boys helped each other.’

Crucially, on the other hand, morphological *reflexives* are prohibited from cooccurring with an overt anaphor, as illustrated by (13) below, making a sharp contrast with the morphological *reciprocal* in (11) above.

⁴ In other words, we do agree with both Bar-Asher Siegal (2016) and LeTourneau (1998) about the need to postulate a phonetically empty anaphor in some cases of morphological reciprocals in Arabic. However, the successful cooccurrence of morphological T-marking with an overt reciprocal anaphor as in (12) makes us believe that T-marking should not be regarded as a morpho-syntactically metamorphosized empty reciprocal anaphor as described in (7).

⁵ When a verb is reciprocalized with morphological T-marking, the internal argument must be accompanied by some designated prepositions like *maʃ* ‘with’ also in JA as shown in (12). In generative syntactic terms, this preposition may be regarded as compensating for the case absorbed by T-morpheme. Whether or not such characterization turns out to be plausible, these prepositions seem to be nearly transparent in semantic terms, only adding somewhat redundant information in a way similar to *on* in *rely on* in English. Since the PP complement overtly expressed in this construction is interpreted as the internal argument, T-morpheme in JA is unlikely to induce detransitivization by “reciprocalization bundling,” contrary to what is proposed for Hebrew by Siloni (2012).

- (13) l-walad t-ʔaddab (*maʕ nafs-uh₁).
 the-boy.NOM T-behaved with self-him.GEN
 ‘The boy behaved himself.’

This observation leads us to take the position that morphologically marked reflexives indeed project only one argument syntactically although we leave it open how such a one-place construction comes to be derived and yields a two-place interpretation without involvement of an empty syntactic argument.

To sum up, comparison of reflexive and reciprocal constructions leads us to the following hypotheses about “morphological” reflexives and reciprocals in JA: (i) T-marked reflexives project only one but T-marked reciprocals project two syntactic arguments, and (ii) when T-marked reciprocals are not accompanied by an overt reciprocal anaphor, they involve a phonetically empty internal argument. These hypotheses thus lead us to shy away from Siloni’s (2001; 2008; 2012) bundling analysis of morphological reciprocals as in (6b), which induces valency reduction, (at least for JA, in which each of the two arguments in the reciprocal construction separately receives a thematic role).⁶

The remainder of this work proceeds as follows. In Sections 2, we first define the terms “collectivity” and “distributivity” as we use in this work. Then, appealing to these notions, we demonstrate the interpretive contrast between “morphological” and “syntactic” reciprocals, which leads us to identify collectivity as the function of the T-morpheme in JA.⁷ Sections 3 attempts to identify the formal apparatuses that capture the semantics of collectivity. Finally in Section 4, we will analyze the morphological properties of the T-marked reflexives and reciprocals, paying attention to the semantics of their input and output verbs.

2. The interpretive properties of reciprocals

In this section, we would like to examine and discuss the interpretive properties of reciprocals in JA. In pursuit of this topic, we will find out that several semantic notions such as collectivity, reciprocity and simultaneity are rather intricately intertwined in the interpretation of reciprocals in JA. Our goal in this section is: (i) to clearly define the notion of collectivity and distinguish it from distributivity, and (ii) to abstract collectivity from reciprocity and simultaneity/symmetry.

⁶ We believe that we can further argue that reflexives and reciprocals should *not* be treated parallelly in the syntax of JA when we closely examine some facts on binding. Such arguments are summarized in Appendix A along with the objection to those arguments raised by one of the anonymous reviewers.

⁷ This claim contrasts with the oft-observed descriptive characterization of this morpheme as a reciprocalizer (Wright 1896; Thackston 2000; Fischer 2002; Holes 2004; Ryding 2005; Fassi Fehri 2012, among others). Despite the existence of such a disparity, we will continue to use “morphological reciprocal” as a convenient label of the T-marked construction, contrasting it with “syntactic reciprocal.”

In the process of fulfilling these tasks, we will be led to identify the exact semantic function associated with the so-called “reciprocal” T-morphemes in JA as *collectivity*.

2.1 Collectivity and distributivity

In Section 1, we presented two types of reciprocal sentences in JA: syntactic and morphological reciprocals. We then adopted the hypothesis that they are structurally parallel and involve a two-place construction even when the complement reciprocal anaphor of the *T-marked* predicate is realized covertly, as exemplified in (14) and (15).

- (14) $l\text{-}\eta\text{axu}_1$ $w\text{-}\eta\text{uxt-uh}_2$ ηaawanu **baŋð-hum** **l-baŋð**_{<1↔2>}.
the-brother.NOM and-sister-his.NOM helped **some-them.ACC** **the-some.ACC**
‘The brother and his sister helped each other.’

- (15) $l\text{-}\eta\text{axu}_1$ $w\text{-}\eta\text{uxt-uh}_2$ $t\text{-}\eta\text{aawanu}$
the-brother.NOM and-sister-his.NOM T-helped
{maŋ **baŋð-hum** **1-baŋð**_{<1↔2>} / [e]_{RECIP<1↔2>} }.
with **some-them.GEN** **the-some.GEN**
‘The brother and his sister helped each other.’

When it comes to their interpretations, however, “syntactic” and “morphological” reciprocals exhibit differences. First, in the syntactic reciprocal sentence as in (14), the expressed reciprocity can be semantically construed either distributively or collectively. That is, the two subeventualities involved in the reciprocal helping in (14) may have “distributively” occurred at different times and/or in different locations, for example, as in the situation such that the brother financially supported the sister while she went to college and the sister looked after the brother when he became seriously ill in later years. Alternatively, the reciprocal helping in (14) may also have occurred “collectively,” with the involved subeventualities significantly overlapping (spatio)temporally, for example, as in the situation such that the brother and the sister clung to each other and warmed themselves in a cabin when they got lost in a snow-covered mountain (with “complete simultaneity”). Collective reciprocal helping in (14) may also have occurred with somewhat looser, “partial (or situational)” simultaneity, for example, in the situation such that the brother and his sister teamed up and collaboratively finished cleaning the house, the brother vacuuming the floor and the sister wiping the windows and furniture. We define the notions collectivity and distributivity in this particular way in the context of the event semantics (as will be formalized in Section 3). Therefore, caution should be exercised to eliminate any misunderstanding that may arise when these terms are used in some other sense.

The availability of both collective and distributive interpretations in (14) can be confirmed when we observe that this sentence can be felicitously followed either by distributive adjuncts as in (16a–b) or by a collective adjunct as in (16c).⁸

- (16) a. **bi-t-tanaawub** ‘taking turns/in turn’
in-the-turn.GEN
- b. **b-fakel** **munfašil** ‘separately’
in-manner.GEN separate.GEN
- c. **sawijjeh** ‘together’
together.ACC

Reciprocal constructions involving morphologically T-marked verbs as in (15), on the other hand, only permit the collective interpretation, and are thus compatible with the collective but not distributive adjuncts, as shown in (17).⁹

- (17) l-ʔaxu₁ w-ʔuxt-uh₂ t-ʔaawanu (maʕ **baʕḏ-hum** **1-baʕḏ**_{<1↔2>})
the-brother.NOM and-sister-his.NOM T-helped with some-them.GEN the-some.GEN
{ ^{ok}**sawijjeh** / [#]**bi-t-tanaawub** }.
together.ACC in-the-turn.GEN
‘The brother and his sister helped each other { ^{ok}together / [#]taking turns }.’

The same contrast arises even when the reciprocal anaphor *baʕḏ-hum 1-baʕḏ* is replaced by an empty anaphor [e]_{RECIP} (indicated in (17) as the optionality of the overt anaphor).¹⁰

⁸ A little caution is called for when we appeal to these adjunct expressions as a litmus test for a distributive-collective contrast. First, while we use the translation ‘in turn’ here and below for naturalness of English translation, *tanaawub*, a term commonly used to indicate “taking separate shifts,” e.g., in a Jordanian military, would not permit any overlap of either complete or partial simultaneity as described above, contrary to English *in turn*, which seems to be compatible with certain type of partial simultaneity.

Second, *sawijjeh* and its English translation ‘together’ can be used as in (i) below to indicate the more abstract sense ‘in collaboration with’ rather than describing the overlap of subeventualities.

(i) Three artists painted this picture **together**.

In our test, we are appealing to *sawijjeh* ‘together’ indicating ‘the significant overlap of subeventualities’ rather than ‘in collaboration with’. We are grateful to an anonymous reviewer for urging us to be alert to such a variation.

⁹ Throughout the paper, we will not indicate functional categories irrelevant to our arguments. Also, we will let the judgments of Arabic sentences be reflected in their English translations as well with *, #, etc., even if those English sentences themselves are grammatical. Various types of extra examples that we were unable to incorporate into the main text are listed in Appendix E.

¹⁰ Siloni (2002: 391; 2012: 266, 282) and Dimitriadis (2008a: 332; 2008b: 379) tried to capture the difference between the “morphological” and the “syntactic” reciprocals in Hebrew and Greek, respectively, in terms of the possible number of events each construction may describe. The generalization we offered above about JA is quite similar to theirs (at least in essence) and provides a further crosslinguistic basis for the generalization in question. We inherited the terms “collectivity” and “distributivity” and their characterization from their works and elaborated them further as described above.

Another such pair of “syntactic” and “morphological” reciprocal sentences are presented in (18) below, in which the same contrast can be observed.

- (18) a. $l\text{-}taa\check{d}green_{<1+2>}$ $faa\check{s}alu$ $ba\check{s}\check{d}\text{-}hum$ $l\text{-}ba\check{s}\check{d}_{<1\leftrightarrow 2>}$.
the-businessmen.DL.NOM haggled **some-them**.ACC **the-some**.ACC
 $\check{s}ala$ $l\text{-}\check{?}as\check{s}aar$ { $^{ok}wa\check{d}\check{h}\text{-}an$ $li\text{-}wa\check{d}\check{h}$ / $^{ok}bi\text{-}t\text{-}tanaawub$ }.
on the-prices.GEN face-ACC for-face.GEN in-the-turn.GEN
‘The two businessmen haggled with each other over the prices { ok face to face / ok in turn }.’
- b. $l\text{-}taa\check{d}green_{<1+2>}$ $t\text{-}faa\check{s}alu$ (ma $\check{s}$ $ba\check{s}\check{d}\text{-}hum$ $l\text{-}ba\check{s}\check{d}_{<1\leftrightarrow 2>}$)
the-businessmen.DL.NOM T-haggled with **some-them**.GEN **the-some**.GEN
 $\check{s}ala$ $l\text{-}\check{?}as\check{s}aar$ { $^{ok}wa\check{d}\check{h}\text{-}an$ $li\text{-}wa\check{d}\check{h}$ / $\#bi\text{-}t\text{-}tanaawub$ }.
on the-prices.GEN face-ACC for-face.GEN in-the-turn.GEN
‘The two businessmen haggled with each other over the prices { ok face to face / $\#$ in turn }.’

Once again, the distributive interpretation available in the syntactic reciprocal in (18a) becomes unavailable in the morphological reciprocal in (18b). Therefore, (18a) is compatible with the situation such that the first businessman haggled with the second businessman over the prices of some items on one occasion and the second businessman haggled with the first businessman over the prices of different items on another occasion. On the other hand, (18b) is compatible only with the situation such that the two businessmen haggled with each other over the prices on a single occasion. Note here that the collectivity in (18b) involves what was identified above as overlapping subeventualities induced by “partial (or situational)” simultaneity. While the mutual haggling took place “collectively” within a single business session, the two businessmen were not necessarily talking to each other at the very same moment. In other words, the overlapping subeventualities required by the T-morpheme in JA can be induced by either complete or partial simultaneity.¹¹

Thus, we now face a task of having to explicate the observed contrast between “syntactic” and “morphological” reciprocals in JA, in particular, why the collective interpretation becomes obligatory only in the latter.

2.2 Reciprocity and collectivity

One might consider that the obligatory collectivity discussed above arises in one way or another from a reciprocal anaphor as its source. This, however, does not seem to be the case since the contrast between “syntactic” and “morphological” reciprocals arises while both of them involve a reciprocal anaphor, for example as we saw in (18a–b) above. Reciprocal anaphors, in other words, are susceptible to both collective and distributive interpretations.

¹¹ Such looser “partial” simultaneity arises also with predicates like $f\text{-}q\text{-}r$ ‘fight’ and $r\text{-}s\text{-}l$ ‘correspond’. In one fighting event, for example, punching by each participant does not necessarily take place at exactly the same moment, exhibiting only partial simultaneity which can still induce collective reciprocity.

We also observe in (19) below that a morphologically T-marked predicate is incompatible with adjunct expressions enforcing distributivity such as *bi-t-tanaawub* ‘taking turns’ and *b-fakel munfaṣil* ‘separately’.

- (19) l-ʔaxu w-ʔuxt-uh **t-ʕaawanu** maʕ ʔabuu-hum
 the-brother.NOM and-sister-his.NOM T-helped with father-their.GEN
 { ^{ok}sawijjeh / #bi-t-tanaawub / #b-fakel **munfaṣil** }.
 together.ACC in-the-turn.GEN in-manner.GEN separate.GEN
 ‘The brother and his sister helped their father {^{ok}together/#taking turns/#separately}.’

Note that (19) does not involve any reciprocal anaphor or reciprocal interpretation.

Researchers like Dimitriadis (2008b), Siloni (2008) and Bar-Asher Siegal (2016) report that morphologically derived predicates similar to the T-marked verb in (19) involve reciprocity between the subject and the (comitative) complement in Greek, German and Hebrew, which they called “discontinuous (reciprocal) construction.” The sentence in (19), however, can make up a felicitous discourse even when it is followed by another sentence negating reciprocity (square-bracketed below) in (20).

- (20) l-ʔaxu w-ʔuxt-uh **t-ʕaawanu** maʕ ʔabuu-hum
 the-brother.NOM and-sister-his.NOM T-helped with father-their.GEN
 [laakin ʔabuu-hum **maa t-ʕaawan** maʕ-hum].
 but father-their.ACC NEG T-helped with-them.GEN
 ‘The brother and his sister helped their father [but their father did **not** help them].’

This suggests that (19) does not involve any reciprocal interpretation between the subject and the complement.¹² After we have discussed the semantics and morphology involved in collectivity, we will point out in Section 4.4 that “discontinuous reciprocals” indeed are possible in JA but only under restricted circumstances.

In short, collectivity is enforced on morphologically T-marked predicates whether or not reciprocal anaphors are introduced into a sentence, i.e., independently of reciprocity. This observation suggests that the function of the T-morpheme in JA should be characterized independently of reciprocity.

2.3 Lexically induced simultaneity and collectivity

Many researchers have identified and discussed what they call “symmetrical” (or “collective/group-level/mutual”) predicates in English as in (21) below, which permit what can be sensed as

¹² We will report the same phenomenon observed in a fair number of other Arabic dialects in Appendix C, where we present the results of our small-scale investigation.

reciprocity without requiring an overt reciprocal anaphor (e.g., Lasnik 1995; Carlson 1998; Link 1998; Peres 1998; Lønning 2011).

(21) John and Mary **kissed/hugged/met/married**.

Crucially, the “reciprocity” expressed by each verb in (21) does not permit distributed multiple events. “John and Mary kissed,” therefore cannot properly describe, for example, the situation such that Mary kissed John on his cheek after John kissed Mary on her forehead. Carlson (1998: 45) argues that “symmetrical” predicates denote a singular event but their lexically entailed mutuality holding between the atomic individuals of a plural subject induces obligatorily reciprocal and collective interpretations. (See also Link 1998: 7, 49 for a similar analysis.) It seems to be true that the lexical meanings of “symmetrical” predicates induce the sense of simultaneity and mutuality because of the necessarily arising shared experience/situation in the induced eventualities.

One may suspect that such lexical semantics of the involved predicates may also play a crucial role in inducing obligatory collectivity in JA, at least in some of the cases discussed above, e.g., (18b) involving the “symmetrical” predicate *faaʃalu* ‘haggled’. It indeed is the case that a distributive adjunct is anomalous in (22) below, in which the “symmetrical” verb *gaabalu* ‘met’ appears even without any reciprocal anaphor present overtly or covertly.

(22) Bader w-Bandar *t-gaabalu* maʃ Sawsan (#**bi-t-tanaawub**).
 Bader.NOM and-Bandar.NOM T-met with Sawsan.GEN in-the-turn.GEN
 ‘Bader and Bandar met Sawsan (**#in turn**).’

There are reasons, however, to believe that such lexically induced simultaneity is not necessarily the source of the obligatory collectivity we observed in JA (although it seems to be inducing plurality of eventualities). First, as Carlson (1998: 43) notes, the predicates in question behave as “collective” only when they are used *intransitively* with a plural NP as their subject. The transitive use of *kiss* in (23a) below, for example, can denote multiple events but no simultaneous or symmetrical entailment arises at least between the atomic individuals of the plural subject *John* and *Mary*.

(23) a. John and Mary *kissed* **their daughter**.
 b. John and Mary *kissed* **each other**.

Even when a reciprocal anaphor appears as the object as in (23b), the transitive use of *kiss* can felicitously describe the distributed multiple events like John’s kissing Mary on the arm and Mary’s kissing John on the arm (observation ascribed to Lila Gleitman). The situation is exactly the same in (24a–b) below, the JA counterpart of (23a–b).

(24) a. Bader w-Sarah *baasu* bint-hum.
 Bader.NOM and-Sara.NOM kissed daughter-their.ACC
 ‘Bader and Sara kissed their daughter.’

- b. Bader₁ w-Sarah₂ **baasu** baʃð-hum l-baʃð_{<1↔2>}.
 Bader.NOM and-Sara.NOM kissed some-them.ACC the-some.ACC
 ‘Bader and Sara kissed each other.’

Note now that the sentence in (22) as well involves a two-place construction, and no simultaneity between *Bader* and *Bandar* seems to be entailed, suggesting that the obligatory collectivity in this sentence (as per our definition) was not caused by lexically induced simultaneity of the verb *gaabalu* ‘met’.

Second, even when an identical “symmetrical” verb is used, whether or not it exhibits obligatory collectivity depends on the form it appears in, i.e., whether it is morphologically T-marked or not. Compare, for instance, (25) below with (22) above.

- (25) Bader w-Bandar **gaabalu** Sawsan (^{ok}**bi-t-tanaawub**).
 Bader.NOM and-Bandar.NOM met Sawsan.ACC in-the-turn.GEN
 ‘Bader and Bandar met Sawsan (^{ok}**in turn**).’

While both sentences involve a symmetrical verb *gaabalu* ‘met’, a distributive reading is permitted in (25) but not in (22). The contrast in question, in other words, is independent of the symmetricity of the verb itself but rather is dependent on the verb forms.

Various other “symmetrical” verbs also exhibit the contrast in question when they appear in “syntactic reciprocal” sentences on the one hand and in “morphological reciprocal” sentences on the other:

- (26) a. z-zalameh₁ w-marat-uh₂ **ʕaanagu** / **baawasu** baʃð-hum l-baʃð_{<1↔2>}
 the-man.NOM and-wife-his.NOM hugged kissed some-them.ACC the-some.ACC
 { ^{ok}**b-lahðet** **bidaajet** **raas** **es-saneh** / ^{ok}**bi-t-tanaawub** }.
 in-moment.GEN beginning.GEN head.GEN the-year.GEN in-the-turn.GEN
 ‘The man and his wife hugged/kissed each other { ^{ok}at the moment the new year began / ^{ok}taking turns }.’
- b. z-zalameh₁ w-marat-uh₂ **t-ʕaanagu** / **t-baawasu** (maʃ baʃð-hum
 the-man.NOM and-wife-his.NOM T-hugged T-kissed with some-them.GEN
 l-baʃð_{<1↔2>}) { ^{ok}**b-lahðet** **bidaajet** **raas** **es-saneh**
 the-some.GEN in-moment.GEN beginning.GEN head.GEN the-year.GEN
 #**bi-t-tanaawub** }.
 in-the-turn.GEN
 ‘The man and his wife hugged/kissed each other { ^{ok}at the moment the new year began / #taking turns }.’

The morphologically unmarked verbs *faanagu* ‘hugged’ and *baawasu* ‘kissed’ in (26a) are compatible with both of the adjuncts ‘at the moment the new year began’ and ‘taking turns’, yielding either a collective or distributive interpretation. When the same verbs are morphologically T-marked as in (26b), on the other hand, the distributive adjunct ‘taking turns’ suddenly becomes incompatible. It is true that the distributive adjunct is incompatible even when the morphologically T-marked verbs in (26b) are not accompanied by an overt reciprocal anaphor (indicated by its optionality). Therefore, they may appear to act as an intransitive “genuine” symmetrical verb at first sight. As discussed in Section 1, however, the seemingly intransitive construction here should be analyzed as involving an empty anaphor $[e]_{\text{RECIP}}$ in the complement, and hence lexically entailed simultaneity is unlikely to arise directly between the conjoined subjects.¹³

Finally, the contrast in question between the T-marked and non-T-marked verbs can be observed even when the verb is non-symmetrical in meaning as in (27) and (28).

- (27) Bader w-Bandar *t-saamaħu* maʃ Sawsan (#*bi-t-tanaawub*).
 Bader.NOM and-Bandar.NOM T-forgave with Sawsan.GEN in-the-turn.GEN
 ‘Bader and Bandar forgave Sawsan (#*in turn*).’

- (28) Bader w-Bandar *saamaħu* Sawsan (^{ok}*bi-t-tanaawub*).
 Bader.NOM and-Bandar.NOM forgave Sawsan.ACC in-the-turn.GEN
 ‘Bader and Bandar forgave Sawsan (^{ok}*in turn*).’

(27) is only compatible with the reading that both Bader and Bandar forgave Sawsan on one occasion while (28) is compatible with not only such a collective reading but also with a distributive reading: Bader and Bandar separately forgave Sawsan, each on a distinct occasion.¹⁴

In short, lexically induced simultaneity of predicates should not be regarded as the only source of the obligatory collectivity observed in morphological reciprocals in JA.

2.4 Collective morpheme

Our investigations so far in this section lead us to identify the exact source of the observed obligatory collectivity. **Table 1**–(i–iii) encapsulate which constructions exhibit optional collectivity and which others obligatory collectivity:

¹³ An empty reciprocal complement is not permitted in the syntactic (i.e., morphologically unmarked) reciprocals as in (26a) in JA. We will discuss this phenomenon in Appendix B when we take up the cross-linguistic variations between Arabic and English.

¹⁴ See also the contrast between (19) and (29) to be discussed shortly below in Section 2.4.

	Optional collectivity		Obligatory collectivity
(i)	(14) + (16)	vs.	(17)
	(18a)	vs.	(18b)
(ii)		vs.	(19)
	(28)	vs.	(27)
(iii)	(25)	vs.	(22)
	(26a)	vs.	(26b)

Table 1: Optional vs. obligatory collectivity.

The contrasts shown in (i) indicate that obligatory collectivity is not induced by reciprocal anaphors, whether they are overt or covert. The observations in (ii) tell us that obligatory collectivity in fact can be observed even when no reciprocity whatsoever is involved in the sentence. Finally, the contrast in (iii) demonstrates that obligatory collectivity arises independently of lexically induced mutuality/simultaneity of the so-called “symmetrical” predicates.

While none of reciprocity, mutuality or simultaneity can induce the obligatory collectivity observed in the data listed on the right-hand side of **Table 1**, there is one thing in common among all of them — the morphological marking of the verbs with the morpheme *t-*. This observation suggests that obligatoriness of collectivity is either directly induced by or at least coincides with the presence of the morpheme *t-*. In fact, as illustrated in (29) below, when we eliminate the T-morpheme from the verb in (19) listed under **Table 1**–(ii), the distributive expression meaning ‘taking turns’ becomes compatible, which further supports this hypothesis. (Note that (29) fills the gap in one of the cells on the left-hand side of **Table 1**–(ii).)

- (29) l-ʔaxu w-ʔuxt-uh ʕaawanu ʔabuu-hum
the-brother.NOM and-sister-his.NOM helped father-their.ACC
{ ^{ok}sawijjeh / ^{ok}bi-t-tanaawub }.
together.ACC in-the-turn.GEN
‘The brother and his sister helped their father { ^{ok}together / ^{ok}taking turns }.’

Thus, while researchers have generally treated the Arabic T-morpheme examined in this section solely as a reciprocal marker, the facts in JA prompt us to reidentify it as a morpheme inducing collectivity rather than reciprocity. In the rest of this work, we would like to pursue this hypothesis and call the T-morpheme in question a “collective morpheme” or “collectivizer.”¹⁵

¹⁵ In Section 4, a more general investigation of the verb morphology in JA will lead us to take the view that the T-morpheme is both *potentially collective* and *potentially reflexive*. In the appendices, we will also explore a possible extension of our proposal on the T-morpheme in JA to English (Appendix B) and other Arabic dialects (Appendix C). It should be made clear, however, that we are *not* denying the existence of reciprocalizing morphemes in other languages.

the two subeventualities. If the subeventualities overlap only in time but not in space, we can simply replace the spatial and temporal function K with the temporal function τ as in (32b). Roughly, a property $\llbracket Coll_{(\text{Spatio})\text{temporal}} \rrbracket^g$, i.e., collectivity, is defined to hold of P if and only if the subeventualities e_1 and e_2 in the eventuality e significantly overlap in time (and space). We need to take heed of the following three points when we adopt this approach. First, when we discuss the notion of collectivity appealing to $Coll$, we are paying attention to the overlapping property of eventualities rather than reciprocity, entailed simultaneity, or a single eventuality achieved by a group. Second, we consider that the presence of $Coll$ in a sentence gives rise to *obligatory* collectivity and also that some constructions may or may not involve $Coll$, which gives rise to optional collectivity. Third, $Coll$ in (32) is concerned with collectivity in the event semantics while Buring's (2005) $Dist$ as in (31) is defined without incorporating event semantics. We therefore need to redefine $Dist$ in (31) in order to combine it with $Coll$ in (32). This adjustment can be made by changing the semantic type of the predicate verb V in (31) from $\langle e, \langle e, t \rangle \rangle$ into $\langle e, \langle e, \langle e, t \rangle \rangle \rangle$ and accordingly the types of constituents that combine with its maximal projection VP in order to avoid any type mismatches. The semantic type of the $Dist$ operator, in other words, needs to be changed from $\langle e, t \rangle$ as in (33a) into $\langle e, \langle e, t \rangle \rangle$ as in (33b). (Here, e is meant to indicate the semantic type of eventuality.)

- (33) a. $\llbracket Dist \rrbracket^g = \lambda P_{\langle e, t \rangle} \lambda Z. \forall x \sqsubseteq_A Z, P(x)$ (= (31))
 b. $\llbracket Dist \rrbracket^g = \lambda P_{\langle e, \langle e, t \rangle \rangle} \lambda Z \lambda e. \forall x \sqsubseteq_A Z, P(x)(e)$

Redefined under the event semantics, the function of $Dist$ as in (33b) now comes to identify the existence of a plural eventuality such that each atomic individual of a plural argument (typically a subject) functions as a participant in respective subeventualities. Note then that the subeventualities introduced this way may take place either collectively or distributively (as per our definitions of these notions) even if the naming of this operator “*Dist*” (inherited from the literature) may give the impression that obligatory distributivity arises. In a sense, this operator is labeled “distributive” only because it distributes the atomic individuals of the plural argument among the subeventualities.

In our analysis, we postulate $Coll$ and $Dist$ as the heads projecting their own phrases $CollP$ (Collective Phrase) and $DistP$ (Distributive Phrase), respectively, above VP as in the syntactic representation indicated in (34).

- (34) $[_{TP} \text{Sbj}_1 [_{CollP} (Coll) [_{DistP} (Dist) [_{VP} t_1 V \text{Complement}]]]]$

Presumably, the introduction of these operators is optional though $Dist$ must select VP, and $Coll$ must select $DistP$ as a complement, once they are introduced.

The sentence (30) above thus can be syntactically represented as in any of (35a–c).

- (35) a. John and Mary [_{vp} sang a song].
 b. John and Mary [_{DistP} **Dist** [_{vp} sang a song]].
 c. John and Mary [_{CollP} **Coll** [_{DistP} **Dist** [_{vp} sang a song]].

Since (35a) does not involve *Dist*, no subeventualities arise, which presumably results in the group reading of the subject involving only a single eventuality. *Dist* in (35b), on the other hand, identifies a plural eventuality, namely, the subeventualities in each of which an atom of the plural subject participates. Since *Coll* is not involved, its subeventualities receive no restriction and can be ambiguously interpreted as either collective or distributive. To the contrary, *Coll* introduced in (35c) imposes obligatory overlap on the involved subeventualities and requires that they take place collectively. We consider that which of these interpretations arises depends on the context/pragmatics, and that, in order to represent each such context-dependent interpretation properly, either or both of *Dist* and *Coll* are introduced into a syntactic representation. The claim we made on the T-morpheme in JA therefore can be formally embodied as its requirement to co-occur with *Coll*.

One important and interesting question is whether and what role *Dist* and *Coll* might play in the interpretation of reciprocals. Reciprocity induced by reciprocal anaphors is known to exhibit compatibility with distinct contexts involving varied “strength” of reciprocity, ranging from the “strong reciprocity” (i.e., reciprocity *conjunctively* holding between the involved atomic individuals) as in (36) below to the “weak (or non-symmetric) reciprocity” (i.e., reciprocity *disjunctively* holding between the involved atomic individuals) as in (37).¹⁷

(36) John and Mary kissed **each other**.

(37) The trays are stacked on top of **each other**.

Different researchers adopt different approaches to cope with the variation of actual interpretations which fall somewhere between these two interpretations. Dalrymple et al. (1998), for example, postulate a small inventory of reciprocal meanings and hypothesize that a sentence takes the strongest meaning in the inventory when it is consistent with the given context. Bar-Asher Siegal (2020), on the other hand, hypothesizes that the weakest reciprocity is the basic meaning of what he calls “NP-strategy” reciprocals, which can be strengthened depending on the given context and intended implicature. Perhaps, even the opposite approach is at least logically possible, which hypothesizes that the strongest reciprocity is the basic meaning and it can be weakened depending on the context and implicature. Since our main interest in this section is the semantics of collectivity, we are not in a position to evaluate or select any of such approaches to capture the interaction between the semantics of reciprocals and their contexts, or to argue for

¹⁷ In Appendix C, we will see that the “collective-distributive ambiguity” can be observed and the T-morpheme eliminates this ambiguity even in “weak (or non-symmetrical) reciprocals” in JA and other dialects of Arabic.

any particular semantic analysis of reciprocity.¹⁸ In Appendix D, nonetheless, we will explore how the operators *Dist* and *Coll* can be utilized in describing the semantic interpretation of collectivity when it is combined with (strong) reciprocity in JA.

4. Aspects of T-morphology in Jordanian Arabic

We now pay attention to morphology. The traditional study of verb morphology in Arabic has recognized two distinct functions of morphological T-marking as summarized in **Table 2**.¹⁹

	(i) Input Verb Form	(ii) Derived Verb Form	(iii) Root	(iv) Derived Meanings
a.	2	5	<i>ḥ-m-m</i> ‘bathe’	<i>t-ḥammamu</i> ‘they bathed{ themselves / #each other }’
b.	3	6	<i>s-m-ḥ</i> ‘forgive’	<i>t-saamaḥu</i> ‘they forgave{ each other / #themselves }’
c.	1	8	<i>r-m-j</i> ‘throw’	<i>ṯir-t-amu</i> ‘they threw { themselves / #each other }’
d.	1	8	<i>f-r-g</i> ‘separate’	<i>ṯif-t-ragu</i> ‘they separated from{ each other / #themselves }’

Table 2: Overview of T-morphology in JA.

First of all, as exemplified in column (iv), the T-morpheme appears at surface as a prefix in Forms 5 and 6 but as an infix in Form 8. This variation, however, is generally considered to be induced by phonology and all instances of T-morpheme are said to be identifiable as a prefix (see McCarthy 1981: 390). Following this well-accepted suggestion, we will analyze all instances of T-morpheme appearing in all of Forms 5, 6 and 8 as prefixes.

The prefix *t-* serves as a reflexivizer in Form 5 (**Table 2**–a) but it has been said to serve as a reciprocalizer in Form 6 (**Table 2**–b). The same T-morpheme in Form 8, on the other hand, has been said to ambiguously serve as a reflexivizer as in **Table 2**–c or as a reciprocalizer as in **Table 2**–d. Reflecting the conclusion we drew in Section 2.4, however, we recognize the T-morpheme in **Table 2**–b and **Table 2**–d as a collectivizer rather than a reciprocalizer with the assumption that reciprocity in collective sentences generally arises due to the syntactic presence of a reciprocal anaphor, which may be overt or covert.²⁰

¹⁸ See also Sabato & Winter (2012) and Mari (2014), among others, for relevant discussions.

¹⁹ See Appendix E for further examples. It is well known that Semitic languages exhibit a quite rich variety of non-concatenative verb morphology involving consonantal roots into which vowels and affixes are slotted for inflection and derivation, and JA is no exception to it. **Table 2** certainly is not comprehensive and by no means attempts to encapsulate all the semantic signification of all verb forms in JA. We are concentrating here on the verb forms and their meanings that are of interest to the main topic of this work. For detailed discussion of Arabic verbal morphology in general, see McCarthy (1981; 1982); Watson (2002); Holes (2004); Ryding (2005; 2014); and Essien (2014), among others.

²⁰ In Section 4.4 below, we will discuss a context in which reciprocity arises with Forms 6 and 8 without involving a reciprocal anaphor.

The segregation of these two distinctive functions of T-marking can be confirmed in (38a–d) below. First, the distributive adjunct ‘taking turns/in turn’ is felicitous in (38a) and (38c) but not in (38b) and (38d). That is, collectivity is optional in the former verb forms (5&8) but obligatory in the latter forms (6&8).

(38) a. Reflexive (Form 5):

el-ʔuxteen ez-zyaar t-ḥammamen bi-l-ḥammaam
the-sisters.DL.NOM the-little.NOM T-bathed in-the-bath.GEN
{ **b-nafs** **el-wagt** / **bi-t-tanaawub** }.
in-same.GEN the-time.GEN in-the-turn.GEN
‘The two little sisters bathed { **themselves** / **#each other** } in the bath
{ **simultaneously** / **taking turns** }.’

b. Collective (Form 6):

l-waladeen t-saamahū { **b-nafs** **el-wagt** / **#bi-t-tanaawub** }.
the-boys.DL.NOM T-forgave in-same.GEN the-time.GEN in-the-turn.GEN
‘The two boys forgave { **each other** / **#themselves** } { **simultaneously** / **#in turn** }.’

c. Reflexive (Form 8):

l-waladeen ʔir-t-amu ʔa-l-ʔasirrah { **b-nafs** **el-wagt** /
the-boys.DL.NOM T-threw on-the-beds.GEN in-same.GEN the-time.GEN
bi-t-tanaawub }.
in-the-turn.GEN
‘The two boys threw { **themselves** / **#each other** } on the beds { **simultaneously** /
taking turns }.’

d. Collective (Form 8):

z-zalameen ʔif-t-ragu { **b-nafs** **el-wagt** / **#bi-t-tanaawub** }.
the-men.DL.NOM T-separated in-same.GEN the-time.GEN in-the-turn.GEN
‘The two men separated from { **each other** / **#themselves** } { **simultaneously** / **#in turn** }.’

Second, as indicated by the translations, reflexive interpretations are available in (38a) and (38c) but prohibited in (38b) and (38d). Reciprocal interpretations, on the other hand, are available in (38b) and (38d) but prohibited in (38a) and (38c). In Sections 1 and 2 above, we adopted the view that T-reflexives in JA make up a one-place construction while T-collectives make up a two-place construction which permits an empty reciprocal anaphor as the complement. All the interpretive restrictions observed in (38) are in accordance with these morpho-syntactic analyses.

How exactly can the T-morpheme exhibit reflexivity in some cases and collectivity in others? Although this issue is independent of other claims we made above and its full-scale pursuit goes beyond the scope of this paper, we would like to conduct an initial survey on this issue. One possible and perhaps reasonable approach is to hypothesize that the T-morpheme is inherently specified as both *potentially reflexive* and *potentially collective*, and that each of these inherent

properties is activated only when some specific properties exist in the verb form the T-morpheme is attached to. Though briefly, we would like to explore an analysis of T-morphology along this line, couching it in the syntactic and semantic analyses argued for above.

As summarized in **Table 2**–(i–ii) above, the T-morpheme in JA takes three distinct verb forms as input and derives three distinct verb forms as output. In what follows, we will examine each such derivation paying attention to the properties of the input verb forms and the specific effects the T-morpheme achieves in each case. For the sake of brevity of our discussion, we will abbreviate “Verb Forms 1, 2, 3, 5, 6, and 8” as “F1, F2, F3, F5, F6 and F8,” respectively, sometimes referring to the verb forms themselves and other times using them as mere labels. In addition, we will use “transitive, unergative and unaccusative” to label the distinct *thematic properties of the original consonantal verb roots* as in **Table 2**–(iii) (rather than characterizing a syntactic construction co-occurring with any verb form).

4.1 Derivation of Verb Form 5 from Verb Form 2

As was introduced in **Table 2**–a above, F5 is derived when the T-morpheme selects F2 as its input. F2, whose form is characterized by the consonant gemination of the second radical ($C_1VC_2C_2VC_3$), makes up a *causative* construction, as illustrated by (39)–(41) below (cf. Wright 1896; Holes 2004; Ryding 2005, among others). (We are suggesting here that F2-unaccusatives should also be regarded as involving causation, extending a traditional view.)

(39) a. **F2-transitive:**

l-ʔustaað ʕallam ʔ-ʕaalib l-ʕazif
the-teacher.M.NOM caused.to.**learn** the-student.M.ACC the-playing.ACC
ʕa-li-bjaanoh
on-the-piano.GEN
‘The male teacher caused the male student to learn (playing) the piano.
(= The male teacher taught the male student (playing) the piano.)’

b. l-ʔabu mallak ʔibn-uh sajjaarah.
the-father.NOM caused.to.**own** son-his.ACC car.ACC
‘The father caused his son to own a car. (= The father got his son a car.)’

(40) **F2-unergative:**

huuh maffa / sabbah / ʔaddab ʔibn-uh.
he.NOM caused.to.**walk** caused.to.**swim** caused.to.**behave** son-his.ACC
‘He caused his son to walk/swim/behave. (= He made his son walk/swim/behave.)’

(41) **F2-unaccusative:**

a. ʔaww eʃ-james fattah li-wruud.
light.NOM the-sun.GEN caused.to.**open** the-flowers.ACC
‘The sunlight caused the flowers to open/bloom.’

- b. huuh kassar ʃ-ʃubbaak.
 he.NOM caused.to.**break** the-window.ACC
 ‘He caused the window to break. (= He broke the window.)’

The effect of F5-T-marking then can be regarded as *requiring the causer and causee in F2 be united in reference*, as illustrated in (42)–(44).

(42) F5-transitive:

- a. ʈ-ʈaalib t-ʃallam l-ʃazif ʃa-li-bjaanoh
 the-student.M.NOM T-caused.to.learn the-playing.ACC on-the-piano.GEN
 ‘**The male student** caused **himself** to learn (playing) the piano.’ (= The male student learned / taught himself / self-taught (playing) the piano.)’
- b. l-ʔibin t-mallak sajjaarah.
 the-son.NOM T-caused.to.own car.ACC
 ‘**The son** caused **himself** to own a car. (= The son got himself a car.)’

(43) F5-unergative:

- l-ʔibin t-maffa / t-sabbah / t-ʔaddab.
 the-son.NOM T-walked T-swam T-caused.to.behave
 ‘**The son** caused **himself** to walk / swim / behave. (= The son walked / swam / behaved.)’

(44) F5-unaccusative:

- a. li-wruud t-fattahat.
 the-flowers.NOM T-caused.to.open
 ‘**The flowers** caused **themselves** to open. (= The flowers bloomed.)’
- b. l-gazaaz t-kassar.
 the-glass.NOM T-caused.to.break
 ‘**The glass** caused **itself** to break. (= The glass broke by itself.)’

Based upon the interpretations of F5-transitive as in (42) and F5-unergative as in (43), F5-T-morpheme has traditionally been characterized as a “reflexivizer,” while F5-unaccusative as in (44) has been treated separately as (medio)passive (Wright 1896; Ryding 2005). Our extension of the causative analysis of F2 to unaccusatives as in (41), however, makes it possible for us to assimilate all cases of F5 in (42)–(44).

In short, F2 makes up a *causative* construction, and T-morpheme selecting F2 *requires the causer and causee in F2 be united in reference*, deriving F5 as a “reflexive” verb.

4.2 Derivation of Verb Form 6 from Verb Form 3

As was summarized in Table 2–b above, an F6 is derived when the T-morpheme selects F3 as its input. F3 ($C_1V_1V_1C_2V_2C_3$) is identified by the vowel lengthening taking place in its input verb form F1. F3 verbs often imply some kind of loose symmetry/parallelism, as in (45a–b).

(45) F3-unergatives:

- a. l-walad ḍaaḥak *(ṣaaḥb-hu).
 the-boy.NOM laughed friend-his.ACC
 ‘The boy laughed with his friend.’
- b. l-walad raakaḍ *(l-bint).
 the-boy.NOM ran the-girl.ACC
 ‘The boy ran with the girl.’

It is rather difficult to formally characterize the exact semantico-syntactic effect achieved by F3. One descriptive generalization we have managed to come up with, however, is that F3 seems to require the agent argument of the input verb to appear with *one more animate* (typically *human*) *participant as the initiator of the same eventuality*, which in effect paves the way for F3 to involve a *plural eventuality* under certain conditions to be discussed shortly below. Moreover, what is noteworthy in (45a–b) is that F3-unergatives must appear with an accusative-marked object which is interpreted as an additional agent of the verb. Putting emphasis more on this interpretive effect than its form, we will label this added participant in examples like (45) “*agentive object*.”²¹

Similarly, F3-transitives must appear with such an “agentive object” when the object of a transitive verb is inanimate while this extra participant is prohibited when the object is animate, as illustrated by the contrast in each of (46)–(47).²² (Note the distinct positions of * between each pair of sentences.)

(46) F3-transitive:

- a. l-walad raahan *(ṣaaḥb-uh) b-maṣaari.
 the-boy.NOM bet friend-his.ACC in-money.GEN
 ‘The boy bet money with his friend. (= Both the boy and his friend bet money.)’
- b. z-zalameh naada *(ṣaaḥb-uh) l-bint.
 the-man.NOM called friend-his.ACC the-girl.ACC
 Intended: ‘The man called (the name of) the girl with his friend. (= Both the man and his friend called (the name of) the girl.)’

(47) F3-transitive:

- a. l-walad ṣarak *(ṣaaḥb-uh) ʔa-l-ʔafa.
 the-boy.NOM shared friend-his.ACC on-the-dinner.GEN
 ‘The boy shared dinner with his friend. (= The boy and his friend shared dinner.)’

²¹ As far as we can tell, there is no evidence that a transitive verb taking an experiencer subject can productively constitute F3, which has led us to decide on the label *agentive object*. Benmamoun (2003: 53–56) characterizes F3 as a “plural form of a verb” and describes the interpretation induced by such an additional agent as “the meaning of sharing in the same activity.” We would like to credit this insight to him, and hope that we have clarified it further.

²² See Benmamoun (2003: 56) for similar examples in Standard Arabic.

- b. l-walad ʕaawan (*ʕaahb-uh) l-bint.
 the-boy.NOM helped friend-his.ACC the-girl.ACC
 Intended: ‘The boy helped the girl with his friend. (= Both the boy and his friend helped the girl.)’

In other words, when (and only when) the agent in the input verb is not accompanied by the second animate/human participant, F3 adds an agentive object, which is to be interpreted as an extra initiator of the same action.

On the other hand, since unaccusative verbs do not involve any agent argument to begin with, they cannot appear in F3 at all, as illustrated in (48) below. Note that these sentences are ungrammatical whether the second participant appears as an accusative-marked object or not, and also whatever interpretive role this participant may be intended to play.

(48) *F3-unaccusatives

- a. *li-zhuur faatahat ({ s-samaad / l-jaasmiin }).
 the-roses.NOM opened the-fertilizer.ACC the-jasmine.ACC
 Intended: ‘The roses bloomed ({ due to the fertilizer / with the jasmines }).’
- b. *l-gazaaz kaasar ({ l-hawa / s-saguf }).
 the-glass.NOM broke the-wind.ACC the-roof.ACC
 Intended: ‘The glass broke ({ by the wind / with the roof }).’

Turning now to F6, we have noted that F6-T-morpheme is attached to only those F3s that can yield a *plural eventuality*. This state of affairs is a natural consequence if F6-T-morpheme is a collectivizer, as argued for in Section 2.4. If only a single eventuality is involved in its input verb form, a collectivizer cannot fulfil its function properly and fails to be interpreted. For instance, while the F3s involving a single eventuality as in (49a–b) below have no problem, the same verbs cannot be turned into F6s, as shown in (50a–b).

- (49) a. z-zalameh naada l-walad b-haðiik l-laḥḏah.
 the-man.NOM called the-boy.ACC in-that.GEN the-moment.GEN
 ‘The man called the (name of the) boy at that moment.’
- b. z-zalameh xaabar ʕ-ʕurṭah ʕan-illi ʕaar.
 the-man.NOM informed the-police.ACC on-what.GEN happened
 ‘The man informed the police about what had happened.’
- (50) a. *z-zalameh t-naada ʕa-l-walad b-haðiik l-laḥḏah.
 the-man.NOM T-called on-the-boy.GEN in-that.GEN the-moment.GEN
 Intended: ‘The man called the (name of the) boy at that moment.’

- b. *z-zalameh t-xaabar ʔa-ʃ-ʃurṭah ʔan-illi ʃaar.
 the-man.NOM T-informed on-the-police.GEN on-what.GEN happened
Intended: ‘The man informed the police about what had happened.’

If the input F3 involves a plural eventuality, on the other hand, the derived F6 becomes legitimate, with its subeventualities interpreted as *obligatorily collective*, as we have confirmed with numerous examples in Sections 2.1–2.3. (See (17), (18b), (19), (27), (22) and (26b) listed in the right column of **Table 1.**)

Having examined all the F6 examples at hand, we have noted that a plural eventuality which their input F3s are required to induce can be established in at least four ways as described in (A)–(D) below.

(A) It is created by *parallel subeventualities* involving two NPs interpreted as agent as in (51) and (52).

(51) F6-unergative:

- a. l-walad t-ḡaahak *(maʃ ʃaahb-uh) xilal muʃaahadet et-tilfizjoon.
 the-boy.NOM T-laughed with friend-his.GEN while watching.GEN the-television.GEN
 ‘The boy laughed with his friend while watching television. (= Both the boy and his friend laughed while watching television.)’
- b. l-walad t-raakaḡ *(maʃ l-bint).
 the-boy.NOM T-ran with the-girl.GEN
 ‘The boy ran with the girl. (= Both the boy and the girl ran.)’

(52) F6-transitive:

- a. l-walad t-raahan *(maʃ ʃaahb-uh) b-maʃaari ʔala sibaag
 the-boy.NOM T-bet with friend-his.GEN in-money.GEN on race.GEN
 el-ḡiṣneh.
 the-horses.GEN
 ‘The boy bet money with his friend on the horse race. (= Both the boy and his friend bet money on the horse race.)’
- b. l-walad t-ḡaasab *(maʃ ʃaahb-uh) ʔala el-waḡbeh.
 the-boy.NOM T-paid with friend-his.GEN on the-meal.GEN
 ‘The boy paid for the meal with his friend. (= Both the boy and his friend paid (their shares of) the money for the meal.)’

In F6-unergative (51a), for example, the addition of the agentive object induces two parallel subeventualities “the boy laughed” and “his friend laughed.” Similarly in the F6-transitives in (52), the second participant is added as an agentive object within the complement PP and induces an extra subeventuality. Thus, in (52a) for example, two parallel subeventualities “the boy bet money” and “his friend bet money” arise.

- b. **l-waladeen** **raakaðu** ***(l-bint)**.
 the-boys.DL.NOM ran the-girl.ACC
 ‘The two boys ran with the girl.’

We would like to claim, however, that the sentences in (55) only *appear* to involve a one-place construction at surface but in reality involve a two-place construction. In particular, we would like to maintain the hypothesis we adopted in Section 1 and assume that the T-morpheme allows an empty reciprocal anaphor $[e]_{\text{RECIP}}$ to appear as the required agentive object in (55), as shown in (55’).

(55’) F6-**unergative** reanalyzed:

- a. **l-waladeen** _{<1+2>} **t-ḏaahaku** $[e]$ _{<1↔2>} { ^{ok}*b-nafs* *el-wagt*
 the-boys.DL.NOM T-laughed in-same.GEN the-time.GEN
 / [#]*bi-t-tanaawub* }.
 in-the-turn.GEN
 ‘The two boys laughed with **each other** { ^{ok}simultaneously / [#]in turn }.’
- b. **l-waladeen** _{<1+2>} **t-raakaðu** $[e]$ _{<1↔2>} { ^{ok}*b-nafs* *el-wagt*
 the-boys.DL.NOM T-ran in-same.GEN the-time.GEN
 / [#]*bi-t-tanaawub* }.
 in-the-turn.GEN
 ‘The two boys ran along with **each other** { ^{ok}simultaneously / [#]in turn }.’

We may consider that parallel subeventualities are also induced in a crisscrossing fashion here by the reciprocity involved in these sentences, in (55’a) for example, between “each boy (A and B)” and “the other boy (B and A).” Since $[e]_{\text{RECIP}}$ must be anaphoric to a plural subject, a singular subject cannot introduce it, and an (overt) agentive object must appear, as we have already seen in (51). This contrast indirectly supports the postulation of $[e]_{\text{RECIP}}$ in (55’). Moreover, this analysis allows us to capture otherwise improbable reciprocal interpretations detectable in the unergative sentences in (55’) as indicated in their translations.

(C) A plural eventuality required by the F6-T-morpheme can also be created by *symmetrical* verbs, as exemplified by (57a–b) below. With such verbs, none of the participants need to be plural in number.

- (57) a. Bader **t-faaṣal** maṣ Bandar ṣala l-ṯasṣaar
 Bader.NOM T-haggled with Bandar.GEN on the-prices.GEN
 { ^{ok}*waḍḥ-an li-waḍḥ* / [#]*bi-t-tanaawub* }.
 face-ACC for-face.GEN in-the-turn.GEN
 ‘Bader haggled with Bandar over the prices { ^{ok}face to face / [#]in turn }.’

- b. Bader **t-saabag** maʃ Bandar { ^{ok}b-nafs l-wagt
 Bader.NOM T-raced with Bandar.GEN in-same.GEN the-time.GEN
 / #bi-t-tanaawub }.
 in-the-turn.GEN
 ‘Bader raced with Bandar { ^{ok}at the same time / #in turn }.’

We will analyze this case in more detail in Section 4.4 below.

(D) In pursuing the source of a plural eventuality in F6, we have encountered some examples that appear to constitute exceptions to our generalizations. Sometimes, F6 seems to be permitted even when none of (A)–(C) above is achieved, as illustrated by (58)–(60).

- (58) z-zalameh **t-baahaθ** bi-l-muwḏuuʃ (#min wagt la-θaani).
 the-man.NOM T-discussed in-the-subject.GEN from time.GEN to-second.GEN
 ‘The man discussed the subject **thoroughly** (#from time to time).’
- (59) z-zalameh **t-laaʃab** bi-l-natiiḏgeh (#min wagt la-θaani).
 the-man.NOM T-manipulated in-the-result.GEN from time.GEN to-second.GEN
 ‘The man manipulated the result **completely** (#from time to time).’
- (60) z-zalameh **t-daaras** bi-l-fikrah (#min wagt la-θaani).
 the-man.NOM T-studied in-the-idea.GEN from time.GEN to-second.GEN
 ‘The man studied the idea **intensely** (#from time to time).’

That is, F6 seems to be permitted even when a sentence does not involve any of plural participants, parallel subeventualities, or symmetrical verbs. None of these examples involves the second human/animate participant, either. The possibility of F6 in such sentences may suggest some essential inadequacy of the generalization made above. By pushing forward some creative and abstract thinking, however, we can envisage a way to capture cases like (58)–(60) under the proposed generalization.

First, the rejection of a distributive adjunct “from time to time” in these sentences suggests that these sentences all involve obligatory collectivity although such an interpretation is not directly or immediately detectable. Instead, they all seem to convey some implied meanings which may be characterizable as certain types of *intensity* or *exhaustiveness* as indicated by “thoroughly,” “completely” and “intensely” in their translations. We would like to claim here that such intensified interpretations are arising because the eventuality involved in each of these sentences can be analyzed as plural, involving *the convergence of multifaceted or multi-angled subeventualities*. Simply put, we consider that the event of “the man’s discussing the subject” in (58), for instance, was “thorough” because he examined various aspects of the subject.

Similarly, “the man’s manipulating the result” in (59) was complete because he maneuvered various significant aspects of the result. As a matter of fact, these sentences can co-occur with the adjuncts indicating a plural eventuality but they cannot co-occur with the adjuncts indicating only a single eventuality, as illustrated in (61a–c).

- (61) a. “The man discussed the subject” in (58) followed by:
 { ^{ok}min ɕɔawaanib kθiireh / #min ɕɔaanib waahad }.
 from sides.GEN many.GEN from side.GEN one.GEN
 ‘{ ^{ok}from many sides / #from one side }’
- b. “The man manipulated the result” in (59) followed by:
 { ^{ok}min ʔiddit ɕɔawaanib / #min ɕɔaanib waahad }.
 from several.GEN sides.GEN from side.GEN one.GEN
 ‘{ ^{ok}from several sides / #from one side }’
- c. “The man studied the idea” in (60) followed by:
 { ^{ok}min ʔiddit zawaaja / #min zaawjeh wahadeh }.
 from several.GEN angles.GEN from angle.GEN one.GEN
 ‘{ ^{ok}from several angles / #from one angle }’

The suggested analysis of (58)–(60) is also compatible with the observation that the verbs which express a one-time event cannot constitute a similar F6, as already seen in (50a–b).

Finally, even if the proposed account of (58)–(60) is adopted, we must still explain how such a plural eventuality can be made possible with the involvement of only a single initiator of the event. To fulfill this task, we would like to appeal to the notion “guise” proposed by Heim (1998) to explain why the violation of Principles B and C of the binding theory can be evaded in examples like (62a–b), respectively (Reinhart 1983; Higginbotham 1985; Grodzinsky & Reinhart 1993).

- (62) a. You know what Mary, Sue and John have in common?
 Mary admires John, Sue admires him, and **John** admires **him** too.
- b. **He** put on **John**’s coat; but only John would do that; so he is John.

In these utterances, the speaker believes, and intends the hearer to believe, that the highlighted NPs are coreferential despite the apparent violation of Principles B and C, respectively, induced by such interpretations. Heim (1998: 214–217) attempts to solve this conundrum by postulating two distinct “guises” of the same referent *John* and *him/he* in the given contexts which would lead to different cognitive values. In (62a), for example, *John* as an admirer and *him* as the admired appear as two distinct guises. Since the two guises are distinct from each other, they do not give rise to a binding relation in syntax, and hence do not violate Principles B/C. In their semantic interpretation, on the other hand, the interlocutors appeal to their knowledge and/or assumption that the two guises share the same referent and interpret them as coreferential.

Returning now to (58)–(60), we have claimed above that a plural eventuality in these examples is induced by the convergence of multifaceted subeventualities. Then, it probably is not entirely inconceivable that multiple guises as the initiators of such multiple subeventualities appear in these examples. In (59), for example, “*the man* as one guise addressed the problem from a theoretical angle,” and “*the man* as another guise addressed the problem from a practical angle,” and so on. Such multiple guises can induce parallel subeventualities while they refer to one and the same participant. The approach adopted in (D) is admittedly exploratory, but it allows us to capture not only the obligatory collectivity but also the otherwise inexplicable source of the intensified interpretation observed in (58)–(60), assimilating them into those discussed in (A)–(C). We will discuss this phenomenon in Section 4.4 below again, which suggests that (D) in fact can be regarded as a quite rational option the grammar of JA permits.

In short, F3 pluralizes an eventuality, requiring the involvement of *more than one human (or animate) participant as the initiator of the same eventuality*. Then, T-morpheme selecting F3 requires the subeventualities involved there to significantly overlap in time (and location), deriving F6 as a “collective” verb.

4.3 Derivation of Verb Form 8 from Verb Form 1

F8 is derived when the T-morpheme selects F1 as its input. F8 ($\text{2iC}_1\text{VC}_2\text{VC}_3$) is derived from F1 ($\text{C}_1\text{VC}_2\text{VC}_3$) by: (i) *t*-prefixation (t-C_1) followed by (ii) Flop (C_1t), and (iii) compensatory 2i -prefixation to avoid both a consonant cluster ($\#C_1\text{t}$) and a vowel ($\#\text{iC}_1\text{t}$) word-initially. (See Wright 1896: 41; McCarthy 1981: 390; among others.) F1 is an inflected but *unaugmented* verb form, which serves as the foundation in deriving many other verb forms. As already described in Table 2–(c–d) above, F8 sometimes yields reflexivity just as F5 does, but at other times it yields obligatory collectivity just as F6 does. How can we rationalize these phenomena? We will attempt to provide an answer below based upon the following reasoning. When F8 yields a reflexive interpretation, it must be the case that the F1 it selects shares whatever aspect of F2 which activates the “potential reflexivity” of the T-morpheme. Likewise, when F8 yields an obligatory collective interpretation, it must be the case that its input F1 shares the aspect of F3 which activates the “potential collectivity” of the T-morpheme.²³ Let us now discuss each of these cases in turn.

First, since the basic function of the T-morpheme as a *reflexivizer* is to require *united reference* of the participants within a single eventuality, the only way the input *unaugmented* verb (F1) can satisfy this requirement would be to provide two participants. This reasoning will lead us

²³ Since these conditions must be met by the F1 verbs’ idiosyncratic lexical properties, only a limited number of F8s are derivable compared to F5s and F6s, which are derivable from more “productively derived” F2s (causativized by gemination) and F3s (pluralized by vowel lengthening).

to consider that F8 necessarily selects a *transitive F1* whose two arguments can enter a felicitous *self-related* (or *intrapersonal*) relation when they are interpreted as united in reference, as exemplified in (63).²⁴

(63) **Reflexive F8:**

- a. l-ʔixwah ʔix-*t*-abu bi-s-siddeh.
the-brothers.NOM T-hid in-the.attic.GEN
‘The brothers hid themselves into the attic.’
- b. li-wlaad ʔir-*t*-amu ʔa-l-ʔasirrah.
the-boys.NOM T-threw on-the-beds.GEN
‘The boys threw themselves on the beds.’

Reflexive F8, on the other hand, is never possible with intransitive F1, i.e., unergative or unaccusative verbs.

In short, the reflexivity in F8 is induced more or less in the same way as that in F5, except that the *united reference* of the participants within a single eventuality is made available by the *unaugmented* verb (F1) which denotes a *self-related* (or *intrapersonal*) relation.²⁵

Next, the T-morpheme as a *collectivizer* requires the involvement of a *plural eventuality* in the input verb. In most, if not all, cases of collective F8s, this requirement seems to be fulfilled when the input unaugmented verb (F1) is *symmetrical* in nature, as in (64a–b).

(64) **Collective F8 (Plural subject):**

- a. z-zalameen_{<1+2>} ʔif-*t*-aragu [e]_{<1↔2>}.
the-men.DL.NOM T-separated
‘The two men separated from each other.’
- b. l-waladeen_{<1+2>} ʔix-*t*-alaṭu [e]_{<1↔2>}.
the-boys.DL.NOM T-mingled
‘The two boys mingled with each other.’

Both of these sentences exhibit only a collective reciprocal interpretation (e.g., (38d)) and their subjects must be plural (as will be shown directly below in (65a–b)). This set of facts naturally leads us to postulate an empty reciprocal anaphor as an agentive object in these sentences. The collectivity of the T-morpheme in (64a–b), in other words, seems to be activated by the plural

²⁴ In Section 1, we observed some facts that led us to believe that T-reflexives are *syntactically* realized as a one-place construction. As expected, these reflexive sentences can exhibit either collective or distributive interpretations.

²⁵ As an anonymous reviewer pointed out to us, there is a puzzling complementarity involved in the reflexive sentences in JA — when F1-reflexivization is possible, F2/F5-reflexivization is not possible. On the other hand, when F2/F5-reflexivization is possible, F1-reflexivization is not possible due to the absence of the corresponding F1 form. The examination of these lexical gaps would have to be left for future research.

eventuality in F1 that is induced perhaps by the combination of (A) (parallel subeventualities), (B) (plural participant) and (C) (symmetrical verbs) discussed in Section 4.2 above.

If we alter the plural subject in (64a–b) into a singular one, on the other hand, an overt agentive object is required to appear (in the PP complement), as in (65a–b).

(65) **Collective F8** (Singular subject):

- a. *z-zalameh* *ʔif-t-arag* **(ʕan marat-uh)*
the-man.NOM T-separated from wife-his.GEN
‘The man separated from his wife.’
- b. *l-walad* *ʔix-t-alaṭ* **(bi-l-bint)*
the-boy.NOM T-mingled in-the-girl.GEN
‘The boy mingled with the girl.’

Here, a plural eventuality was induced by the addition of an agentive object combined with a symmetrical verb ((C)). The situation is similar to (A) but the involved subeventualities are reciprocal rather than parallel. We will discuss this case in more detail in Section 4.4 below.

Thus, the collectivity in F8 is induced more or less in the same way as that in F6, except that (C) seems to be always involved for F8. We have not encountered any collective F8 licensed by (D) (convergence of multifaceted subeventualities), either. As will be discussed directly below in Section 4.4, however, there is a good reason for these restrictions.

4.4 Collectivity with single subjects

Having examined both semantics and morphology of collectivity, we now are ready to rationalize how a plural eventuality is induced with a singular subject in some T-marked collective constructions.

Since the T-morpheme in JA is a collectivizer, it needs to be associated with the operator *Coll*. *Coll* in turn needs a plural eventuality to operate on, and hence requires the presence of *Dist*, as in (66) (cf. (34)).

(66) $[_{TP} \text{Sbj}_1 [_{CollP} \text{Coll} [_{DistP} \text{Dist} [_{VP} t_1 \text{T-V Complement}]]]]$

When the subject is singular and *Dist* cannot identify a plural eventuality in the sentence, the grammar of JA seems to permit two ways to devise a plural eventuality.

One method is to introduce an agentive object and appeal to the symmetrical property of the involved verb. For instance, as we saw with (51) and (52) in Section 4.2, when the F3 (as the input to the T-marking of F6) implies loose symmetricity/parallelism, the introduction of an agentive object makes parallel subeventualities available, which can be collectivized by F6. When the F3 verbs imply more direct symmetricity (i.e., simultaneity), e.g., *race*, *hug*, *kiss*, and *fight*,

the introduction of an agentive object gives rise to a crisscrossing plural eventuality and hence reciprocity. The interpretation derived by this strategy as in (67) below is generally recognized as “discontinuous reciprocity.”²⁶

- (67) el-walad *t-saabag* maʕ ʔuxt-uh.
 the-boy.NOM T-raced with sister-his.GEN.
 ‘The boy raced with his sister.’

JA permits another method to devise a plural eventuality with a singular subject. When the input F3 does not imply any kind of symmetricity/parallelism, e.g., *help*, *forgive* and *blame*, the strategy adopted for (51)–(52) and (67) is not available. We believe that a plural eventuality can be devised in such cases with the “guises” of a singular subject, provided that the emerging multifaceted subeventualities are semantico-pragmatically felicitous, as observed with (58)–(60) in Section 4.2. The same strategy is adopted with an asymmetrical F3 as in (68) below, which cannot induce “discontinuous reciprocity.”

- (68) el-walad *t-ʕaawan* maʕ ʔuxt-uh.
 the-boy.NOM T-helped with sister-his.GEN
 ‘The boy helped his sister **a great deal / in many aspects**.’

The multifaceted subeventualities arising here can be detected by the “intensified” interpretation (*a great deal/in many aspects*) which suddenly pops up in this sentence. The presence of “discontinuous reciprocity” in (67) and its absence in (68) can also be demonstrated when we see that the reciprocity of the former cannot be negated as in (69a) while the reciprocity of the latter can be negated as in (69b).

- (69) a. (67) + #w-laakin ʔuxt-uh *maa t-saabagat* maʕ-uh.
 and-but sister-his.NOM NEG T-raced with-him.GEN
 ‘The boy raced with his sister, #**but** his sister did **not** race with him.’
 b. (68) + ^{ok}w-laakin ʔuxt-uh *maa t-ʕaawanat* maʕ-uh.
 and-but sister-his.NOM NEG T-helped with-him.GEN
 ‘The boy helped with his sister, ^{ok}**but** his sister did **not** help him.’

Note that (69a) also makes a sharp contrast with (20) in Section 2.2, which involves a plural subject.

²⁶ See Dimitriadis (2008a; 2008b) and Siloni (2008) for relevant discussions on the role of symmetricity in discontinuous reciprocity. As far as we can tell, F1 as the input to the collective F8 is always symmetrical in nature in JA. F8 cooccurring with a singular subject therefore always exhibits “discontinuous reciprocity,” as we saw with (65) in Section 4.3.

4.5 Two functions of t-morpheme

The investigation so far has led us to conclude that *potential reflexivity* of the T-morpheme is activated by the input verb's capability of involving an *intrapersonal* (or *self-related*) *relation*, and its *potential collectivity* is activated by a *plural eventuality* the input verb induces in a sentence.

We have also confirmed that each of such activation is clearly segregated between F5 (for reflexivity) and F6 (for collectivity), but both types of activation can take place in F8. One reasonable interpretation of this state of affairs is that the T-morpheme induces reflexivity when it selects the input verb form in which the required united reference can be satisfied, whether it is F1 (intrapersonal transitive) or F2 (causative). Likewise, the T-morpheme induces obligatory collectivity when it selects the input verb form which can supply a plural eventuality, whether it is F1 (symmetrical transitive) or F3 (unergative or transitive involving two human/animate participants). Then, in a sense, F5, F6 and F8 can be regarded as mere descriptive labels for each of such subcases of activation, as summarized in **Table 3**.²⁷

	Activated potential property of T-morpheme	Activation by		Descriptive label
a.	reflexivity	intrapersonal relation in	Causative (F2)	Form 5
b.			Transitive (F1)	Form 8
c.	obligatory collectivity	plural eventuality in	Unergative or Transitive involving two human/animate participants (F3)	Form 6
d.			Symmetric Transitive (F1)	Form 8

Table 3: T-morpheme activation.

Finally, as we have confirmed in **Table 2** at the outset of this section, the two potential functions of the T-morpheme as a reflexivizer and a collectivizer are *exclusive* — if one is possible, the other is not possible. Why do they have to be? The answer comes from the very nature of these two functions. First, reflexivity arises because the T-morpheme requires *united reference* between the two participants in the eventuality. On the other hand, the obligatory collectivity is induced by the T-morpheme when a plural eventuality is made available by the input verb. If the two participants in a plural eventuality are united in reference, the two subeventualities involved there amount to be induced by one and the same participant, that is, in a singular rather than plural eventuality. The potential collectivity of the T-morpheme therefore would fail to be activated. We thus are led to the conclusion that the two participants in the input subeventualities must be

²⁷ See also Doron (2003) for the attempt to unify reflexives and reciprocals paying attention to the distinct number of involved participants.

disjoint in reference in order for the potential collectivity of the T-morpheme to be activated. Since the T-morpheme's reflexivizing function induces united reference and its collectivizing function is realized only with disjoint reference, they are mutually exclusive.

The analysis of verb forms in JA we summarized above is far from being comprehensive and the view pursued in this section is undoubtedly preliminary. We believe, however, that they deserve further scrutiny both by those who find them reasonable and by those who do not, so that we can further advance a formal investigation of the syntactic and semantic aspects of verb morphology in JA, which is only scarcely available in the literature. We also hope that they have a potential to shed light on the semantics of the morphological verbal templates in other Semitic languages as well.

5. Summary and conclusion

This paper started with the observations leading us to believe that morphologically marked *reflexives* in Jordanian Arabic (JA) are one-place predicates, but their *reciprocal* counterparts are two-place predicates, with their internal arguments being realized either covertly or overtly in syntax.

It was then argued that, at least in JA, what has traditionally been regarded as a “reciprocalizing” function of the T-morpheme should be reanalyzed as its “collectivizing” function and that the associated reciprocity arises as the side effects of plural eventualities required by the T-morpheme, i.e., parallel subeventualities holding between two separate participants, plural participants or guises. Though briefly, a formal semantic analysis was also provided for the JA sentences which yield distributivity, collectivity and reciprocity in their interpretation.

In the final section, it was also conjectured how the T-morpheme interacts with the semantico-syntactic properties of the input verb forms and derive distinct verb forms.

Abbreviations

ACC = Accusative

DIST = Distributive

DL = Dual

F = Feminine

GEN = Genitive

JA = Jordanian Arabic

M = Masculine

NEG = Negation

NOM = Nominative

OM = Object marker

RECIP = Reciprocal

REFL = Reflexive

Supplementary files

The supplementary materials for this article can be found here: <https://doi.org/10.16995/glossa.5799.s1>. They contain the following materials:

- **Appendix A:** The structural properties of reciprocals
- **Appendix B:** Possible cross-linguistic investigations of Arabic and English
- **Appendix C:** T-morphemes in various Arabic dialects — An initial investigation
- **Appendix D:** Interaction of collectivity and reciprocity
- **Appendix E:** Further examples of sentences and T-marked verbs

These appendices (paginated as pp. 50-88) are incorporated into this file and start on the next page.

Acknowledgements

We are extremely grateful to: our informants for their patient and sincere participation in our investigations, Tom Grano for his help on semantics, and three anonymous reviewers for their helpful comments and criticisms. We would also like to thank the editor for his patient and sensible treatment of the long and winding review process of this work. The usual disclaimers apply to whatever mistakes and deficiencies originated from our ignorance.

Competing interests

The authors have no competing interests to declare.

7. Appendices:.....	50
Appendix A — The structural properties of reciprocals:	50
Appendix A i — Anaphor binding in reciprocals:	51
Appendix A ii — Reviewer’s objection:.....	57
Appendix A iii — Resolving some potential issues on binding:	64
Appendix B — Possible cross-linguistic investigations of Arabic and English:	68
Appendix C — T-MORPHEMES in various Arabic dialects — An initial investigation:.....	76
Appendix C i — Distributivity with T-MORPHEMES:.....	76
Appendix C ii — Distributivity in non-symmetrical reciprocals:	79
Appendix C iii — Discontinuous reciprocity:	81
Appendix D — Interaction of collectivity and reciprocity:.....	83
Appendix E — Further examples of sentences and T-MARKED verbs:.....	88

7. Appendices:

Appendix A — The structural properties of reciprocals:

In Section 1, we followed Bar-Asher Siegal (2016) and LeTourneau (1998) and adopted the hypothesis that morphologically T-marked reciprocals in JA *always* project *two* (i.e., both external and internal) syntactic arguments even when the internal argument is not realized overtly, as illustrated in (70) (= (15)).

- (70) l-ʔaxu₁ w-ʔuxt-uh₂ t-ʕaawanu
the-brother.NOM and-sister-his.NOM T-helped
{ maʕ baʕð-hum l-baʕð<1↔2> / [e]_{RECIP<1↔2>} }.
with some-them.GEN the-some.GEN
‘The brother and his sister helped each other.’

In the earlier draft of this paper, we presented syntactic support for this hypothesis based upon our observations on anaphor binding in JA. We pointed out that our syntactic arguments nicely dovetail with the semantic motivation provided by Bar-Asher Siegal and support the postulation of an empty reciprocal anaphor. One of the anonymous reviewers, however, questioned the legitimacy of the data we presented and insisted on the rejection of the paper. We therefore have decided to withdraw the section featuring the binding arguments because, first, there already are enough arguments in the field to support the postulation of an empty reciprocal anaphor in Semitic languages, and second, it is not our core proposal. (We appealed to an implicit reciprocal anaphor to capture only one phenomenon — when the generally required agentive object of Form 6 and Form 8 T-marked predicates is missing and at the same time obligatory collectivity is detected, as in (26b) (Section 2.3), (55') (Section 4.2), and (64) (Section 4.3). There, we argued that the postulation of an implicit reciprocal anaphor makes available the plural eventualities required by the T-morpheme as a collectivizer, assimilating the examples in question to other general cases. It provided, so to speak, the last piece to complete our arguments for implicit

anaphors.)

In the remainder of this appendix, we will summarize our binding arguments in Appendix A i, the reviewer’s objection in Appendix A ii, and discuss some potential issues involved in our binding arguments in Appendix A iii.

Appendix A i — Anaphor binding in reciprocals:

Support for the analysis postulating an empty reciprocal anaphor in the morphologically T-marked reciprocals comes when we examine binding. First, we confirm the preliminary facts on anaphor binding in JA, which proceeds in four steps. Let us start with the examination of a *one-place* construction whose verbs do *not* involve morphologically marked reflexive or reciprocal forms. In such a construction in (71a–b) below, a SELF-anaphor introduced in an adjunct (**Ajct**) phrase *cannot be bound by the subject*.

- (71) a. **l-walad₁** *ḡihik* [Ajct baʕid bahdalet { *ṣhaab-uh* / ***nafs-uh₁** }].
 the-boy.NOM laughed after disparagement.GEN friends-his.GEN self-him.GEN
 ‘The boy laughed after the disparagement of {his friends /*himself}.’
- b. **l-mudiir₁** *ṣimil* bi-dʒiddiyeh [Ajct baʕid taʕdʒiif
 the-manager.M.NOM worked in-seriousness.GEN after encouragement.GEN
 { l-muwaḡḡafiin / ***nafs-uh₁** } ʕala f-fuʕul].
 the-employees.GEN self-him.GEN on the-work.GEN
 ‘The (male) manager worked hard after the encouragement of { the employees /
 *himself } for work.’

In (71a), for example, when the nominal in the adjunct *ṣhaab-uh* ‘his friends’ is replaced by the SELF-anaphor *nafs-uh* ‘himself’, the sentence becomes ungrammatical, apparently violating Principle A of the binding theory (Chomsky 1981), or some other locality constraint, when this anaphor attempts to refer to the subject *l-walad* ‘the boy’. (We will examine and reject the analysis postulating a local PRO subject within NP in our examples in **Appendix A iii** below, where we will also discuss the issues concerning c-command.)

The second preliminary observation is that, in a *two-place* construction without *morphological T-marking* as in (72a–b) below, the same restriction is observed but with an additional fact emerging. That is, the SELF-anaphor in the adjunct continues to be prohibited from being bound by the subject but now is *permitted to be bound by the object NP*.

- (72) a. *l-waladeen*₁ *ʕaawanu* ***l-bint***₂ [_{Ajct} *ʕala* *taṭwiir*
the-boys.DL.NOM helped the-girl.ACC on improvement.GEN
{ **nafs-hum*₁ / ***nafs-ha***₂ }].
self-them.GEN self-her.GEN
‘The two boys helped the girl with the improvement of { *themselves / herself }.’
- b. *s-saḍjiineen*₁ *waatu* ***Zeed***₂
the-inmates.M.DL.NOM colluded *Zeed*.ACC
[_{Ajct} *ʕala* *tahriib* { **nafs-hum*₁ / ***nafs-uh***₂ } *min* *s-siḍḍin*].
on decampment.GEN self-them.GEN self-him.GEN from the-jail.GEN
‘The two male inmates colluded with Zeed on the decampment of
{ *themselves / himself } from jail.’

In (72a), for example, the anaphor *nafs-hum* ‘themselves’ cannot be bound by the subject *l-waladeen* ‘the two boys’ but *nafs-ha* ‘herself’ can be bound by the object *l-bint* ‘the girl’. We have observed, in other words, that the binding of the SELF-anaphors in the adjuncts in the *two-place* construction is oriented toward the *objects* rather than the subjects in sentences like (72a–b) in JA.²⁸

In the third step of preliminary investigation, we witness that the same binding facts are observed when the *two-place verbs* are *morphologically T-marked* and select a PP complement, as in (73a–b).²⁹

²⁸ How this state of affairs arises is an issue we cannot pursue in this work. See Antonenko (2012), for example, for relevant discussion on how the subject/object-orientation of anaphor binding is achieved in syntax.

²⁹ Morphological T-marking per se in these sentences does not induce reflexivization or reciprocalization. Recall our discussion on the way its semantic function is determined in Section 4.

- (73) a. l-waladeen₁ *t*-ʕaawanu **maʕ** l-bint₂ [A_{jet} ʕala taṭwiir
the-boys.DL.NOM T-helped with the-girl.GEN on improvement.GEN
{ *nafs-hum₁ / **nafs-ha**₂ }]
self-them.GEN self-her.GEN
‘The two boys helped the girl with the improvement of { *themselves / herself }.’
- b. s-sadʒiineen₁ *t*-waaṭu **maʕ** Zeed₂
the-inmates.M.DL.NOM T-colluded with Zeed.GEN
[A_{jet} ʕala tahriib { *nafs-hum₁ / **nafs-uh**₂ } min s-sidʒin].
on decampment.GEN self-them.GEN self-him.GEN from the-jail.GEN
‘The two male inmates colluded with Zeed on the decampment of { *themselves / himself } from jail.’

Therefore, the restriction in a two-place construction imposed on the binding of *SELF-anaphors* in adjuncts should perhaps be described as their “complement-orientation.” The readers should note here that we are NOT claiming that anaphors in JA in general cannot refer to the subject. We are certainly aware that a *SELF-anaphor appearing as an object is subject-oriented* as seen in (1) in Section 1. It, however, is not a matter of concern to us. We are reporting cases in which *anaphors appearing in adjuncts refer to the object of a preposition/verb rather than a subject*.

The forth and the final preliminary observation is that, when the verbs in (72) and (73) select an *overt reciprocal anaphor* as their complement, the SELF-anaphor in the adjunct can naturally be oriented toward this complement, whether the verb is morphologically unmarked as in (74) or marked as in (75).

- (74) a. l-waladeen<1+2> ʕaawanu **baʕḏ-hum** **l-baʕḏ**<1↔2>
the-boys.DL.NOM helped some-them.ACC the-some.ACC
[A_{jet} ʕala taṭwiir **nafs-hum**<1+2>].
on improvement.GEN self-them.GEN
‘The two boys helped each other with the improvement of themselves.’
- b. s-sadʒiineen<1+2> waaṭu **baʕḏ-hum** **l-baʕḏ**<1↔2>
the-inmates.M.DL.NOM colluded some-them.ACC the-some.ACC
[A_{jet} ʕala tahriib **nafs-hum**<1+2> min s-sidʒin].
on decampment.GEN self-them.GEN from the-jail.GEN
‘The two inmates colluded with each other on the decampment of themselves from jail.’

(75) a. l-waladeen<1+2> *t*-ʕaawanu maʕ **baʕð-hum** **l-baʕð**<1↔2>
the-boys.DL.NOM T-helped with some-them.GEN the-some.GEN

[_{Ajct} ʕala taṭwiir **nafs-hum**<1+2>].
on improvement.GEN self-them.GEN

‘The two boys helped each other with the improvement of themselves.’

b. s-sadʕiineen<1+2> *t*-waaṭu maʕ **baʕð-hum** **l-baʕð**<1↔2>
the-inmates.M.DL.NOM T-colluded with some-them.ACC the-some.ACC

[_{Ajct} ʕala tahriib **nafs-hum**<1+2> min s-sidʕin].
on decampment.GEN self-them.GEN from the-jail.GEN

‘The two inmates colluded with each other on the decampment of themselves from jail.’

Note that this complement orientation is masked because the complement anaphor in turn is bound by the subject in each example. The facts observed in (72) and (73), however, urge us to consider that the SELF-anaphor *nafs-hum* ‘themselves’ in all of (74a–b) and (75a–b) becomes acceptable because it can be bound by the complement *baʕð-hum l-baʕð* ‘each other’ rather than the subject.

Now, crucially to us, when the *overt* complement anaphors are *eliminated* from (75a–b), the SELF-anaphor in the adjunct can still be interpreted as referring back to the subject:

(76) a. l-waladeen<1+2> *t*-ʕaawanu [**e**]<1↔2> [_{Ajct} ʕala taṭwiir **nafs-hum**<1+2>].
the-boys.DL.NOM T-helped on improvement.GEN self-them.GEN

‘The two boys helped each other with the improvement of themselves.’

b. s-sadʕiineen<1+2> *t*-waaṭu [**e**]<1↔2>
the-inmates.M.DL.NOM T-colluded

[_{Ajct} ʕala tahriib **nafs-hum**<1+2> min s-sidʕin].
on decampment.GEN self-them.GEN from the-jail.GEN

‘The two inmates colluded with each other on the decampment of themselves from jail.’

If “morphological” reciprocals as in (76a–b) are analyzed as involving a one-place construction, their successful binding would remain mysterious, given that the SELF-anaphors in question are prohibited from being directly bound by the subject in a one-place construction, as we saw in (71). If, on the other hand, these sentences are analyzed as involving a phonetically empty

anaphor in a two-place construction as indicated in (76a–b), the facts can be captured straightforwardly, being assimilated to what we saw in (74) and (75).

Note further that the reciprocals in (76a–b) (now analyzed as involving a two-place construction) make a sharp contrast with morphologically T-marked *reflexives* as in (77a–b) below, which project only one argument syntactically. (cf. (13) in Section 1)

- (77) a. **l-walad₁** **t-ʔaddab** [A_{ject} baʕid taʔniib { ʃhaab-uh / ***nafs-uh₁** }].
 the-boy.NOM T-behaved after reprimand.GEN friends-his.GEN **self-him.GEN**
 ‘The boy behaved himself after the reprimand of { his friends / *himself }.’
- b. **l-walad₁** **t-ħammam** [A_{ject} la-muraaḏaat { ʔumm-uh / ***nafs-uh₁** }].
 the-boy.NOM T-bathed for-satisfaction.GEN mother-his.GEN self-him.GEN
 ‘The boy bathed himself for the satisfaction of { his mother / *himself }.’

In (77b) just above, for example, the nominal in the adjunct *ʔumm-uh* ‘his mother’ cannot be replaced by the SELF-anaphor *nafs-uh* ‘himself’.

The contrast between one-place and two-place constructions can also be demonstrated in the opposite way when reflexivization is established *syntactically* as in (78a–b) below, where a transitive verb selects a reflexive anaphor as its complement.

- (78) a. l-walad₁ ʔaddab **nafs-uh₁**
 the-boy.NOM behaved self-him.ACC
 [A_{ject} baʕid taʔniib { ʃhaab-uh / ^{ok}**nafs-uh₁** }].
 after reprimand.GEN friends-his.GEN self-him.GEN
 ‘The boy behaved himself after the reprimand of { his friends / ^{ok}himself }.’
- b. l-walad₁ ħammam **nafs-uh₁** [A_{ject} la-muraaḏaat
 the-boy.NOM bathed self-him.ACC for-satisfaction.GEN
 { ʔumm-uh / ^{ok}**nafs-uh₁** }].
 mother-his.GEN self-him.GEN
 ‘The boy bathed himself for the satisfaction of { his mother / ^{ok}himself }.’

Note that the SELF-anaphor in the adjunct can now refer back to the subject, presumably because it is successfully bound by the object in accordance with the “complement orientation.” This object in turn is bound by the subject, letting the SELF-anaphor in the adjunct refer to the

subject.

To sum up, close examination of binding in reciprocal constructions supports our hypotheses about “morphological” reciprocals: **(i)** T-marked reciprocals always project two syntactic arguments, and **(ii)** when T-marked reciprocals are not accompanied by an overt reciprocal anaphor, they involve a phonetically empty internal argument.

Table 4 describes the informants we consulted with in order to confirm the correctness of the first author’s judgment on the JA examples presented directly above (and elsewhere in this paper). All investigations were conducted in the form of a questionnaire followed by post-hoc solicitation of additional comments. The stimulus sentences were presented with relevant discourses and/or contexts, and all of the judgments provided here unanimously supported our predictions.

Table 4: Informants in the initial investigation of binding in JA.

Speaker	Number	Gender	Age	Number of linguists involved
Jordanian	At least 4, sometimes 5	2 females • 3 males	20s - 1 30s - 2 40s - 1 50s - 1	1

Appendix A ii — Reviewer’s objection:

One of the anonymous reviewers questioned the credibility of the data supporting the complement-orientation of a SELF-anaphor in JA discussed above. S/he did not directly evaluate the examples we presented, describing them as “quite distracting” but instead provided what s/he described as “so naturally occurring” Twitter messages in (79)–(81) below (clipped from a larger discourse and rewritten in JA).

(79) ...**(pro)** tʃaawanau maʃ el-ʃaduw ʔidd **nafs-hum**.
(**pro**) T-cooperated with the-enemy against **self-them**
‘**They** (= the Palestinians) cooperated with the enemy against **themselves**.’

(80) ...**(pro)** tʃaawanu maʃ eʃ-ʃahaajneh ʔidd **nafs-hum**...
(**pro**) T-cooperated with the-Zionists.GEN against **self-them**
‘**They** (= the Palestinians) cooperated with the Zionists against **themselves**.’

(81) ...**ʃibjaan eθ-θawrah we-l-ʔixwaan** tʔaamaru maʃ el-ʃaskar
fighters the-revolution and-the-brothers T-conspired with the-soldiers
ʔidd **nafs-hum**.
against **self-them**
‘**The fighters of the revolution and the brothers**³⁰ conspired with the soldiers
against **themselves**.’

All of the original Twitter messages involved Standard (literary) Arabic perhaps mixed with each Tweeter’s own (spoken) dialect. The reviewer claimed that the anaphor in these examples can be bound by the subject rather than the object, and hence provides direct counterexamples to “the complement-orientation of an anaphor from within the adjunct PP,” which we reported on some JA sentences. (Recall that, in our binding arguments, we did NOT deny the subject-orientation of an object anaphor in JA.)

We rewrote those Twitter examples into JA and had them checked by 19 Jordanian

³⁰ “The brothers” here refer to the members of the Islamic brotherhood party.

speakers in total.³¹ These informants all reported that the anaphor in each of these examples cannot refer to the subject. Although our paper was written with the intention of presenting empirical and theoretical studies strictly on Jordanian Arabic, we decided to extend our investigation from JA to other Arabic dialects, though in a very small scale, since we surmised that this discrepancy of judgment may have arisen from some dialectal variations we were not aware of. Again, we rewrote (79)–(81) in four distinct dialects and had them examined by one each speaker of Kuwaiti, Saudi, Egyptian and Palestinian Arabic. The result we obtained was the same — the anaphor binding by the subject was rejected by all of those informants.

Moreover, we obtained one interesting observation in our post-hoc survey. One informant pointed out to us that, if these sentences were ever accepted (though marginally), the SELF-anaphors are required not to refer to the subject but to some different people. For example, (79) and (80) would have to be interpreted as “They (= the Palestinian betrayers) cooperated with the enemy/Zionists **against their** (= the Palestinian betrayers’) **fellow Palestinians.**” Likewise, (81) would have to be interpreted as “The (betraying) fighters of the revolution and the brothers conspired with the government’s soldiers **against their** (= the betraying fighters’) **fellow fighters and brothers.**” Upon receiving this input, we catechized the other informants and received a unanimous reaction that they can also stretch their interpretation along this line if they endeavor to. (Some had the impression that such extended interpretation might be easier in Standard Arabic than in their respective dialects.) In other words, there seems to be a way to permit these sentences marginally while the usual sentence-internal anaphor binding is still prohibited in a fair number of dialects in Arabic. The Tweeters in question might have found

³¹ No discourse or context was added in our investigation of the Twitter examples.

nafs-hum ‘themselves’ permissible for the intended readings because the subjects’ deeds were directed against **their** (= the subjects’) **fellow** people, which involves a nuance of reflexivity. See Carroll (1986) for relevant discussions on such marked uses of reflexives, which she calls “non-reflexive anaphors.”

We also believe that (79)–(81) involve the pragmatic difficulty with the complement orientation of the anaphor binding. It is rather difficult to imagine natural pragmatic situations in which “A cooperates/conspires with B against B” makes sense (especially when A is Muslims and B is Zionists). We surmise that this interpretive restriction may have encouraged some Tweeters to bend the grammatical requirement for the complement orientation of anaphor binding.

Table 5 describes the informants involved in our investigations of the Twitter examples (79)–(81).

Table 5: Informants in the investigation of the Twitter examples.

Speaker	Number	Gender	Age	Number of linguists involved
Jordanian	19	2 females • 17 males	20s - 15 30s - 2 40s - 1 50s - 1	2
Saudi • Palestinian • Kuwaiti • Egyptian	1 each, totaling 4	3 females • 1 male	30s - 4	2

The reviewer, who describes her/his own dialect as “close to the Palestinian dialect,” reacted to our responses above as follows. S/he finds the Twitter examples “perfectly grammatical,” in which “the anaphor refers to the subject. It may never refer to the object of preposition; a personal pronoun would be used for that.” S/he also states “the rules of anaphora, at least as pertaining to the topic of the paper, are not different in Standard Arabic compared to the dialects in question ... or shall I say my dialect,” and also “I am not aware of cross-dialectal

variation that would target such structures, and I doubt there is one.” The editor of *Glossa* also solicited judgments on the Twitter examples (79)–(81) from the three syntacticians the reviewer suggested and found that “two of them, Saudi and Jordanian respectively, answered that they did indeed share the judgments of the reviewer.”

Finally, we decided to invite four other Arabic linguists anew to examine some key data (82)–(85) below, hoping that such judgments made by linguists can be regarded equally as trustworthy as those provided by the reviewer’s fellow linguists. **Table 6** describes the new group of linguists we consulted with, and the questionnaire appearing at the end of this appendix (**Appendix A ii**) shows the way the data were collected:

Table 6: Linguists as informants in the investigation of binding in (82)–(85).

Speaker’s Arabic dialect	Number	Gender	Age	Number of linguists involved
Jordanian • Palestinian • Egyptian • Saudi	1 each, totaling 4	2 females • 2 males	20s - 1 30s - 2 40s - 1	4

- (82) l-waladeen tʃaawanu maʃ l-bint₁ ʃala taṭwiir nafs-ha₁.
the-boys.DL.NOM helped with the-girl.GEN on improvement.GEN self-her.GEN

‘The two boys helped **the girl**₁ with the improvement of **herself**₁.’

- (83) a. muḥaawalet ʔintiḥar eṭ-ṭaalib el-maḏḏnuun
attempt.NOM suicide.GEN the-student.GEN the-hazed.GEN
galabat **zummalaaʔ fariig-uh**₁ ḏidd nafs-hum₁.
turned mates.ACC team-his.GEN against selves-them.GEN
‘An attempted suicide of the hazed student turned **his teammates**₁ against **themselves**₁.’
≈ ‘An attempted suicide of the hazed student made **his teammates**₁ blame **themselves**₁.’

- b. eθ-θarθrah galabat el-ḏjiiraan₁ ḏidd baʃḏ-hum l-baʃḏ₁.
the-gossip.NOM turned the-neighbors.ACC against some-them.GEN the-some.GE
‘**The gossip** turned the **neighbors**₁ against **each other**₁.’

- (84) al-filastiniijuun taʃaawanu maʃ aʃ-ʃahjuun₁ ḏidd-uh₁.
the-Palestinians.NOM cooperated with the-Zionist.GEN against-him.GEN
‘The Palestinians cooperated with **the Zionist**₁ against **him**.’

- (85) el-ʔustaaḏ tsaahal maʕ eṭ-ṭaalibeh₁ didd-ha₁.
the-teacher.M.NOM acted-leniently with the-student.F.GEN against-her.GEN
‘The male teacher acted leniently with **the female student** against **her**.’
≈ ‘The male teacher acted leniently toward **the female student**₁, which would not do **her**₁ any good in the end.’

In (82), it was asked if *nafs-ha* ‘herself’ (= *an anaphor located in an adjunct PP*) can refer back to the object *l-bint* ‘the girl’, of course without coindexation indicated. This in fact is identical to (72a) discussed in Appendix A i.

With (83a–b), we examined, in a similar way, if such complement-orientated anaphor binding becomes obtainable even in the sentence structure identical to that in the reviewer’s Twitter examples when pragmatics is properly controlled. It was examined, in other words, if both of reflexive and reciprocal anaphors can comfortably refer back to the object and yield naturally interpreted sentences.

In (84) and (85), it was examined if a pronoun instead of an anaphor indeed would be used to refer back to the object in the construction similar to that in the reviewer’s Twitter examples, as the reviewer claims.

The results are summarized in **Table 7** together with the first author’s judgments. The judgments were made on the binding interpretations indicated by the coindexation in each example.

Table 7: The results of the investigation of binding in (82)–(85).

Speaker’s Arabic dialect	Example				
	(82)	(83a)	(83b)	(84)	(85)
Jordanian (1st author)	ok	ok	ok	*	*
Jordanian	ok	ok	ok	*	*
Palestinian	ok	ok	ok	*	*
Egyptian	ok	ok	ok	*	*
Saudi	ok	*	ok	ok	*

We believe that the results here present a rather clear trend toward the support of our judgments and claims rather than those asserted by the reviewer — predominantly, the anaphor binding in (82) and (83a-b) was permitted and the pronominal binding in (84) and (85) was prohibited.

We presented above what we believe is the faithful record of the disputes over the binding data we presented in Appendix A i. Since the main trend of the judgments provided by our informants was not necessarily shared by some speakers including some Twitter users, one reviewer, and two other linguists, we may indeed be witnessing some hitherto unnoticed variation among Arabic speakers. While we had to withdraw our binding arguments from the main part of the paper, we feel that they should be submitted to the readers together with the reviewer’s objection and our extended investigations so that the readers should be able to make their own judgments on this dispute. The following is the questionnaire used in the extended investigations reported just above.

Questionnaire

Please select one or more of (i)–(iv) for each sentence below. The word forms in the sentences can be changed in accordance with the dialect you speak:

Your dialect	(82)	(83a)	(83b)	(84)	(85)

(82) l-waladeen tʃaawanu maʃ **l-bint** ʃala taṭwiir **nafs-ha**.
the-boys.DL.NOM helped with **the-girl**.GEN on improvement.GEN **self-her**.GEN

‘The two boys helped **the girl** with the improvement of **herself**.’

⇒ Can **nafs-ha** ‘herself’ refer to **l-bint** ‘the girl’?

(i) ___ yes (ii) ___ no (iii) ___ can't tell (iv) ___ too complex sentence structure

- (83) a. muhaawalet ?intiħar eṭ-ṭaalib el-maḏnuun
 attempt.NOM suicide.GEN the-student.GEN the-hazed.GEN
 galabat **zummalaa?** **fariig-uh** ḍidd **nafs-hum**.
 turned mates.ACC team-his.GEN against selves-them.GEN
 ‘An attempted suicide of the hazed student turned his **teammates** against **themselves**.’
 ≈ ‘An attempted suicide of the hazed student made his teammates blame themselves.’
 ⇒ Can **nafs-hum** ‘herself’ refer to **zummalaa? fariig-uh** ‘his teammates’?
 (i) ___ yes (ii) ___ no (iii) ___ can't tell (iv) ___ too complex sentence structure
- b. eṭ-ṭarṭarah galabat **el-ḍjiiraan** ḍidd **baṣṭ-hum** **1-baṣṭ**.
 the-gossip.NOM turned **the-neighbors**.ACC against some-them.GEN the-some.GEN
 ‘The gossip turned the **neighbors** against **each other**.’
 ⇒ Can **baṣṭ-hum 1-baṣṭ** ‘each other’ refer to **el-ḍjiiraan** ‘the neighbors’?
 (i) ___ yes (ii) ___ no (iii) ___ can't tell (iv) ___ too complex sentence structure
- (84) al-filaṣṭiniyyun taṣaawanu maṣ **aṣ-ṣahjuuni** ḍidd-**uh**.
 the-Palestinians.NOM cooperated with **the-Zionist**.GEN against-**him**.GEN
 ‘The Palestinians cooperated with the **Zionist** against **him**.’
 ⇒ Can **-uh** ‘him’ refer to **aṣ-ṣahjuuni** ‘the Zionist’?
 (i) ___ yes (ii) ___ no (iii) ___ can't tell (iv) ___ nonsense sentence
- (85) el-ḥustaaḍ tsaahal maṣ **eṭ-ṭaalibeh** ḍidd-**ha**.
 the-teacher.M.NOM acted-leniently with the-student.F.GEN against-her.GEN
 ‘The male teacher acted leniently with **the female student** against **her**.’
 ≈ ‘The male teacher acted leniently toward **the female student**, which would not do **her** any good in the end.’
 ⇒ Can **-ha** ‘her’ refer to **eṭ-ṭaalibeh** ‘the student’?
 (i) ___ yes (ii) ___ no (iii) ___ can't tell (iv) ___ too complex sentence structure

Appendix A iii — Resolving some potential issues on binding:

There are two potential issues that need to be addressed in relation to the binding arguments presented in Appendix A i above. First, one may adopt Chomsky’s (1986: 123, 167) “PRO in NP” analysis and offer an alternative account to our empty anaphor approach to the “complement orientation.” Under this approach, the successful binding of the SELF-anaphor in the reciprocal sentence (76a) will be ascribed to the presence of a subject-controlled PRO within the adjunct as in (86) below rather than the presence of the empty anaphor in the complement position.

- (86) l-waladeen<1+2> *t*-ḡaawanu [A_{ject} ḡala [NP **PRO**<1+2> tatwiir **nafs-hum**<1+2>]].
the-boys.DL.NOM T-helped on improvement.GEN self-them.GEN
‘The two boys helped each other with the improvement of themselves.’

There are reasons, however, not to adopt this alternative approach. First, it would make incorrect predictions about a similar anaphor binding in one-place constructions. If a subject-controlled PRO were postulated as in (87) below, the SELF-anaphor in the adjunct would be incorrectly permitted. (Recall the examples in (71) above demonstrating the failure of such binding in one-place constructions.)

- (87) l-mudiir₁ *ḡimil* bi-dʒiddiyeh [A_{ject} baʕid
the-manager.M.NOM worked in-seriousness.GEN after
[NP **PRO**₁ taʕdʒiiʕ { l-muwaḏḏafiin / ***nafs-uh**₁ } ḡala ʃ-ʃuyul]].
encouragement.GEN the-employees.GEN **self-him**.GEN on the-work.GEN
‘The manager worked hard after the encouragement of { the employees / *himself } for work.’

The same problem arises in one-place constructions involving “morphological” reflexivization as in (88) (and also in (77b) above when it is reanalyzed with PRO₁ in NP).

- (88) l-walad₁ *t*-ʔaddab [A_{ject} baʕid [NP **PRO**₁ taʔniib { ʃhaab-uh / ***nafs-uh**₁ }].
the-boy.NOM T-behaved after reprimand.GEN friends-his.GEN **self-him**.GEN
‘The boy behaved himself after the reprimand of { his friends / *himself }.’

In fact, if the “PRO in NP” approach is extended to the analysis of two-place constructions, it will end up supporting the postulation of an empty complement anaphor in morphological

reciprocals as in (76a). Recall, first, the binding facts observed in the transitive construction (72a) above (repeated here as (89) with the PRO in NP analysis).

- (89) *l-waladeen*₁ *ʒaawanu* *l-bint*₂
the-boys.DL.NOM helped the-girl.ACC
[_{Ajct} *ʒala* [_{NP} **PRO**₂ *taṭwiir* { **nafs-hum*₁ / ^{ok}*nafs-ha*₂ }].
on improvement.GEN self-them.GEN self-her.GEN
‘The two boys helped the girl with the improvement of { *themselves / ^{ok}herself }.’

Crucially, the SELF-anaphor in the adjunct can be bound by the object *l-bint* ‘the girl’ but not by the subject *l-waladeen* ‘the boys’. If we attempt to capture this contrast by postulating PRO as in (89), we must consider that the PRO here is controlled by the object rather than the subject, contrary to what was assumed in (86).

Furthermore, observe the morphologically T-marked two-place construction in (90).

- (90) *l-muraahiqeen*₁ *t-ʒaaraku* *maʒ-uh*₂ [_{Ajct} *b*-[_{NP} **PRO**_{1/2} *taxliis*
the-adolescents.M.DL.NOM T-joined with-him.GEN in extrication.GEN
{ ^{ok}*ʔuxt-uh* / **nafs-hum*₁ / ^{ok}*nafs-uh*₂ } min *l-maʒaakil*]].
sister-his.GEN self-them.GEN self-him.GEN from the-problems.GEN
‘The two (male) adolescents joined him for the extrication of { ^{ok}his sister / *themselves / ^{ok}himself } from the problems.’

First, when the nominal in the adjunct is *ʔuxt-uh* ‘his sister’, this sentence is grammatical with either the reading (i) “the adolescents₁ worked with him on *their*₁ extrication of his sister from the problems” or (ii) “the adolescents worked with him₂ on *his*₂ extrication of his sister from the problems.” Therefore, if we were to capture this ambiguity with the postulation of PRO as shown in (90), we would have to consider that it can be controlled either by the subject or by the complement. When we replace *ʔuxt-uh* ‘his sister’ with SELF-anaphors in (90), however, a contrast similar to that observed in (89) arises. That is, *nafs-hum* ‘themselves’ referring back to the subject is prohibited while *nafs-uh* ‘himself’ referring back to the complement is permitted despite the ambiguous control, which must be assumed to capture both of the interpretations (i)

and (ii) described just above. This indicates that we should not rely on the postulation of PRO to capture either the “complement orientation” of the anaphor binding (with *nafs-uh* ‘himself’) or the prohibition against its “subject orientation” (with *nafs-hum* ‘themselves’) observed in (90).

If, as in (91) below, we eliminate the overt complement *maʕ-uh* “with him” from (90), we are taken back to the seemingly one-place “morphological reciprocalization” parallel to (76a).

- (91) l-muraahiqaen]_{<1+2>} t-ʃaarakuu [e]_{<1↔2>}
the-adolescents.M.DL.NOM T-joined
[Ajct b- [NP taxliis ^{ok}nafs-hum]_{<1+2>} min l-maʃaakil]].
in extrication.GEN self-them.GEN from the-problems.GEN
‘The two (male) adolescents joined/worked with each other in/on the extrication of
themselves from the problems.’

Unlike in (90), we get the impression that *nafs-hum* ‘themselves’ in (91) can be directly bound by the subject. However, this impression is likely to be delusive, first, since we cannot rely on the postulation of PRO to capture the binding facts in this construction, and second, since the SELF-anaphor in the adjunct cannot be subject-oriented. It therefore seems quite natural and reasonable to postulate the phonetically empty reciprocal anaphor in the complement position and let the SELF-anaphor be bound by it as shown in (91) to rationalize this misleading impression.

The second issue related to our binding arguments is more recalcitrant. We saw above that the reciprocal anaphor in the complement of a morphologically T-marked predicate can serve as the antecedent of a SELF-anaphor located in the adjunct phrase following it. As repeated in (92) below, we further argued that this is what happens whether the anaphor is realized overtly or covertly.

- (92) *l-waladeen*_{<1+2>} [v' *t-ʕaawanu* { *maʕ* ***baʕð-hum*** ***l-baʕð***_{<1↔2>} / [*e*]_{<1↔2>} }]
the-boys.DL.NOM T-helped with some-them.GEN the-some.GEN
[_{Ajet} *ʕala taṭwiir* ***nafs-hum***_{<1+2>}]. (= (75a) + (76a))
on improvement.GEN self-them.GEN
‘The two boys helped each other with the improvement of themselves.’

Here, despite the successful binding of the SELF-anaphor, it apparently is not c-commanded by its antecedent in the complement position. The situation is surely disturbing since c-command has widely been assumed to be a structural condition imposed on binding. It should be noted, however, that this is not a problem peculiar to the “morphological” reciprocal construction we have examined. The same issue arises in a regular (non-reciprocal) transitive construction as in (93) below, in which the SELF-anaphor within the adjunct must be bound by the object NP *l-bint* ‘the girl’ located within V’, again apparently without being c-commanded.

- (93) ***l-waladeen***₁ [v' *ʕaawanu* ***l-bint***₂]
the-boys.DL.NOM helped the-girl.ACC
[_{Ajet} *ʕala taṭwiir* { ****nafs-hum***₁ / ^{ok}***nafs-ha***₂ }] .
on improvement.GEN self-them.GEN self-her.GEN
‘The two boys helped the girl with the improvement of { **themselves* / ^{ok}*herself* }.’

In fact, similar cases involving “binding without c-command” have been widely reported on English. Some cases involve anaphor binding, as in (94) below, and some other cases involve pronominal binding by quantified antecedents, as in (95).

- (94) a. God [gave **the lovers**_{<1+2>}] peace in **each other**’s_{<1↔2>} arms.
b. I talked [_{PP} with **the neighbors**_{<1+2>}] about **each other**_{<1↔2>}.
(Reinhart 1983: 132, 176–7; see also Pollard & Sag 1992, Varaschin 2020)

- (95) a. Our staff keeps a watchful eye [on **every situation**₁] and on **its**₁ developments.
b. [A friend of **each contestant**₁] stood behind **her**₁. (Barker 2012: 621, 623)

When the antecedent is a possessor within a nominal phrase, further complications arise. First, apparently, a contrast arises between the binding of reflexives and reciprocals:

- (96) a. *[**Siegfried₁**'s mother] adores **himself_i**. (Reinhart 1983: 178)
 b. [**The men's_{<1+2>}** books] viciously attacked **each other_{<1↔2>}**. (Hornstein 2001: 219)

Second, anaphors and pronouns are reported to behave differently when they are to be bound by a quantified antecedent as a possessor:

- (97) a. [**Everyone₁**'s father] thinks **he₁**'s a genius. (Higginbotham 1980: 691)
 b. *[**Every girl₁**'s father] admires **herself_i**. (Kayne 1994: 25)

As Barker (2012) summarizes in his overview of the cases involving quantificational binding, the accounts of the puzzles proposed in the literature are quite varied. Some try to adjust syntactic structures or appeal to a mechanism different from binding in order to maintain the validity of the c-command requirement,³² while others try to redefine the notion c-command or simply reject it as a requirement on binding. Unfortunately, pursuit of the solution goes beyond the scope of this work and we must leave it to the future research of ourselves as well as others.

Appendix B — Possible cross-linguistic investigations of Arabic and English:

When we overview our core observations of the JA data and compare them with those of their English counterparts, a grain of an interesting cross-linguistic picture emerges. Although its full-scale pursuit must await future research, we would like to briefly summarize our preliminary sketch of this future topic in this appendix.

Let us begin with the comparison of the JA sentences in the paradigm (98)–(100).

- (98) z-zalameh₁ w-marat-uh₂ baawasu/ʕaawanu **baʕṣ-ḥum** **l-baʕṣ_{<1↔2>}**.
 the-man.NOM and-wife-his.NOM kissed helped some-them.ACC the-some.ACC
 { ^{ok}b-laḥṣet bidaajet raas es-saneḥ / ^{ok}bi-t-tanaawub }.
 in-moment.GEN beginning.GEN head.GEN the-year.GEN in-the-turn.GEN
 ‘The man and his wife kissed/helped each other { ^{ok}at the moment the new year began /
 ^{ok}in turn }.’

³² One may, for example, appeal to the movement of either or both of the anaphor and its antecedent to the periphery of various functional projections over traditional phrases like NP, VP and even PP.

- (99) *z-zalameh₁ w-marat-uh₂ baawasu/ʕaawanu [e]_{RECIP<1↔2>}.
 the-man.NOM and-wife-his.NOM kissed helped

Intended: ‘The man and his wife kissed/helped (each other).’

- (100) z-zalameh₁ w-marat-uh₂ t-baawasu / t-ʕaawanu
 the-man.NOM and-wife-his.NOM T-kissed T-helped
 { maʕ baʕð-hum 1-baʕð<1↔2> / [e]_{RECIP<1↔2>} } (#bi-t-tanaawub).
 with some-them.GEN the-some.GEN in-the-turn.GEN
 ‘The man and his wife kissed/helped each other (#in turn).’

First, we have observed earlier that when an overt reciprocal anaphor accompanies a *non-T-marked* predicate (F3) as in (98), the sentence is compatible with either a collective or distributive adjunct. On the other hand, when an overt anaphor is eliminated from the same sentence as in (99), we now see that it not only disallows the reciprocal interpretation but also makes the sentence *ungrammatical*. However, as we have also observed earlier, the same predicates permit an empty anaphor [e]_{RECIP} when they are morphologically T-marked as in (100), which exhibits *obligatory collectivity* and rejects a distributive adjunct.

If the analyses argued for in the present work are correct, this amounts to the observation that an empty anaphor [e]_{RECIP} in JA can be licensed *only when the collective T-morpheme is present* in the sentence. Note also that this observation holds for both *baawasu* ‘kissed’ and *ʕaawanu* ‘helped’, that is, whether the predicate is the so-called “symmetrical” predicate (which involves lexically induced simultaneity) or “asymmetrical” predicate.

With these observations of JA in mind, let us now turn to similar examples in English as in (101) and (102).

- (101) a. They<1+2> *kissed/hugged* **each other**<1↔2>.
 b. They<1+2> *kissed/hugged* _____<1↔2>.
 (102) a. They<1+2> *helped* **each other**<1↔2>.
 b. They<1+2> *helped* _____ *<1↔2>.

Recall first that, as shown in (101a–b), when “symmetrical” verbs like *kiss/hug* are involved, a reciprocal interpretation is available *with or without* the presence of an overt anaphor *each other*. On the other hand, as Carlson (1998: 43) observed, a reciprocal interpretation is not available when the sentence involves an “*asymmetrical*” verb like *help* but lacks the overt anaphor *each other*, as in (102b).

Second, let us also elaborate on Link’s (1998: 49) original observation and demonstrate that the *absence* of an overt reciprocal anaphor *each other* prohibits a distributive interpretation and requires a collective interpretation of reciprocity even with “symmetrical” verbs as in (101b). This point can be illustrated by the interpretive contrast between (101a) and (101b) as follows. First, all sentences in (101) are grammatical and they are all equally compatible with the adjunct which would prompt *collective reciprocity* in these sentences, for example with “the moment the clock struck 12 midnight.”

The contrast in question shows up, however, between (101a) (with an overt reciprocal anaphor) and (101b) (without it) in the distinct availability of *distributive reciprocity* if such a reading is enforced by a distributive adjunct. If, for instance, the adjunct “in turn” shows up in (101a), it permits the interpretation “John kissed/hugged Mary first, and then Mary kissed/hugged John.” “In turn” showing up in (101b), on the other hand, prohibits such distributive reciprocity but enforces the interpretation “John kissed/hugged someone/something first, and then Mary kissed/hugged (most likely the same) someone/something,” which *does not involve reciprocity*. In short, reciprocity can be achieved either distributively or collectively when an overt reciprocal anaphor is involved as in (101a). On the other hand, reciprocity can be achieved only *collectively* in the same sentence if an overt anaphor is missing, as in (101b). The observation crucial to us here is that *collectivity comes to be imposed obligatorily on a*

reciprocal sentence only when an overt reciprocal anaphor is missing. To the best of our knowledge, Carlson’s observation on (102b) and Link’s observations on (101b) have remained unrelated to this date.

Moreover, let us point out that the observations of Arabic ((98)–(100)) and those of English ((101)–(102)) just presented are intertwined in a puzzling way, as schematically illustrated here:

(103) Arabic (JA: Translated into English):

- (98a') They_{<1+2>} *kissed* **each other** (^{ok}in turn). (≈ (101a') below)
- (98b') They_{<1+2>} *helped* **each other** (^{ok}in turn). (≈ (102a') below)
- (99a') *They_{<1+2>} *kissed* [**e**]_{RECIP<1↔2>}. (≠ (101b') below)
- (99b') *They_{<1+2>} *helped* [**e**]_{RECIP<1↔2>}. (≈ (102b') below)
- (100a') They_{<1+2>} *t-kissed* [**e**]_{RECIP<1↔2>} ([#]in turn). (≈ (101b') below)
- (100b') They_{<1+2>} *t-helped* [**e**]_{RECIP<1↔2>} ([#]in turn). (≠ (102b') below)

(104) English:

- (101a') They_{<1+2>} *kissed* **each other**_{<1↔2>} (^{ok}in turn). (≈ (98'a) above)
- (101b') They_{<1+2>} *kissed* ______{<1↔2>} ([#]in turn). (≈ (100'a)) but (≠ (99'a) above)
- (102a') They_{<1+2>} *helped* **each other**_{<1↔2>} (^{ok}in turn). (≈ (98'b) above)
- (102b') They_{<1+2>} *helped* _____^{*}_{<1↔2>}. (≠ (100'b)) but (≈ (99'b) above)

The two languages are quite parallel when an overt reciprocal anaphor appears in the sentences ((98a–b') ≈ (101a')/(102a')). They are, however, both similar and distinct in intriguing ways when an overt reciprocal anaphor *does not* show up. First, rather unexpectedly, the English symmetrical verb *kiss* in (101b') behaves parallelly to the Arabic *t-marked kiss* in (100a'), while it contrasts with the Arabic “plain” *kiss* in (99'a). When it comes to the “plain” asymmetrical verb *help*, on the other hand, it behaves parallelly in the English sentence (102b') and the Arabic sentence (99'b), both disallowing a reciprocal interpretation.

This enigma, however, can be untangled if our analyses of JA turn out to be correct and reflect more extensive crosslinguistic generalizations as follows. Suppose that a collective

morpheme is not a language particular entity, but it exists more widely in natural languages with its core properties as summarized in (105).³³

(105) a. “ $\mu_{\text{COLL}} \rightarrow$ obligatory collectivity”:

A collective bound morpheme (henceforth, μ_{COLL}) indicates *significant overlap of subeventualities* expressed by a sentence, i.e., *obligatory collectivity* involved there.

b. “ $\mu_{\text{COLL}} \rightarrow V_{\text{plural eventuality}}$ ”:

μ_{COLL} is licensed by an input verb that can yield a *plural eventuality*.

c. “[e]_{RECIP} $\rightarrow \mu_{\text{COLL}}$ ”:

μ_{COLL} must appear in a sentence in order for an empty reciprocal anaphor [e]_{RECIP} to be licensed.

A variety of facts we observed in JA above can be ascribed to these properties of μ_{COLL} , which is t_{COLL} - in case of JA. (105a) accounts for the obligatory collectivity in (100a–b') (contrary to (98a–b')). (105b) prescribes that F6 be derived from F3 and collective F8 be derived from “symmetrical” F1. (105c) induces the ungrammaticality of (99a–b') (contrary to (100a–b')).

Furthermore, let us now explore the possibility that a certain amount of variation is permitted for the realization of a collective morpheme (μ_{COLL}) in different languages. First, let us assume that the morphological realization of μ_{COLL} is cross-linguistically varied, including the possibility that it may be realized as a *phonetically empty* collective morpheme $\emptyset_{\text{COLL}}$. Second, we assume that the exact semantic property of a predicate that licenses μ_{COLL} as in (105b) may vary in different languages within a reasonable range. Such cross-linguistic variations in the grammar concerning morpho-lexical matters are quite often observed and not out of the question, we believe. When we adopt this general view of collective morphemes

³³ Collective morphemes have been verified noticeably often in Oceanic languages (Dench 1995; Dixon 1988, among others). The reciprocal suffix *-aw* in Japanese also seems to exhibit obligatory collectivity together with reciprocity, sometimes involving parallel eventualities.

Pursuing such crosslinguistic generalizations, let us suppose now that a collective morpheme μ_{COLL} exists not only in Arabic but also in English, the crucial difference between the two languages being that it is morphologically realized as t_{COLL} - in JA but as the phonetically empty $-\emptyset_{\text{COLL}}$ in English. This analysis will allow us to reanalyze (101b') above as in (106) below, assimilating it to the Arabic F6 as in (100'a).

(106) They_{<1+2>} *kissed*- $\emptyset_{\text{COLL}}$ [e]_{RECIP<1↔2>} (#in turn). (= (101b'))

Then we can immediately explain why a reciprocal interpretation is available even without the presence of the overt anaphor *each other* in (101b'). Because of the appearance of $-\emptyset_{\text{COLL}}$, an empty reciprocal anaphor [e]_{RECIP} may be introduced in accordance with (105c). Even more importantly, we can also explain why collectivity suddenly becomes obligatory when the overt reciprocal anaphor is eliminated (= Link's observation). The presence of $-\emptyset_{\text{COLL}}$ (interacting with the operator *Coll*) imposes obligatory collectivity in (101b') in accordance with (105a).

Why is a reciprocal interpretation not available in (102b') in English, in which *each other* is missing, contrary to (101b') (= Carlson's observation)? Obviously, the analysis as in (107) below is not available for this case, contrary to (106) above.

(107) *They_{<1+2>} *helped*- $\emptyset_{\text{COLL}}$ [e]_{RECIP<1↔2>}.

We consider that this contrast arises because of the licensing property of $-\emptyset_{\text{COLL}}$ in English such that it must be affixed to a “symmetrical” verb (“ $\mu_{\text{COLL}} \rightarrow V_{\text{symmetric}}$ ”). This permits the representation as in (106) but not that in (107). This may appear to be an arbitrary deviation from the general licensing characterization of μ_{COLL} stated in (105b). When we note that symmetrical eventualities in fact make up a subset of plural eventualities, as illustrated in **Figure 1**, we can say that the licensing property of μ_{COLL} in English ($-\emptyset_{\text{COLL}}$) is only somewhat more narrowly specified than that for general cases.

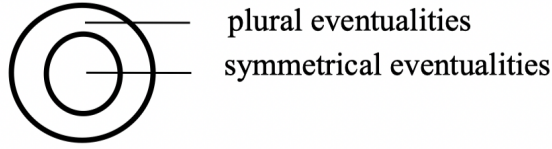


Figure 1: Symmetrical eventualities as a subset of plural eventualities.

From this variation, it follows that “symmetrical” verbs and “asymmetrical verbs” behave differently in English ((101b') vs. (102b')) while they do not in JA ((100'a) $\approx$ (100'b)).

Table 8 depicts the variations of the collective morpheme μ_{COLL} between JA and English.

Table 8: Collective morpheme μ_{COLL} — JA vs. English.

	JA	English
μ_{COLL}	t_{COLL} - (no $\emptyset_{\text{COLL}}$ -)	$-\emptyset_{\text{COLL}}$
Licensing property in the input verb	plural eventuality	<i>symmetrical</i> plural eventuality

Note especially that JA does not have an empty μ_{COLL} ($\emptyset_{\text{COLL}}$ -). This means that an empty reciprocal anaphor $[e]_{\text{RECIP}}$ is not permitted with *non-T-marked* verbs, as shown in (99'a–b). The sentence therefore fails to make up a two-place construction, whether the verb is symmetrical (*baawasu* ‘kissed’) or asymmetrical (*ʒaawanu* ‘helped’), and becomes ungrammatical. To the contrary, *T-marking* and therefore $[e]_{\text{RECIP}}$ as well, is permitted for either type of verbs inducing a plural eventuality, as shown in (100'a–b). This also makes a contrast with the English pair (101b') and (102b'), in which $-\emptyset_{\text{COLL}}$ and hence $[e]_{\text{RECIP}}$ can be licensed only by the *symmetrical* verbs like *kiss* and *hug*.³⁴

³⁴ In Appendix A i, we appealed to a binding test to demonstrate the presence of $[e]_{\text{RECIP}}$ in the T-marked reciprocals in JA (see (76)). We have noted, however, that the same test, unfortunately, will not be credible in the attempt to detect $[e]_{\text{RECIP}}$ in a sentence like (106) in English. Given that a reflexive anaphor in the adjunct in (ia) below can be bound either by the subject or the complement, the successful binding of the anaphor in (ib) does not

To sum up, it was argued in this appendix that otherwise puzzling similarity and discrepancy observed between JA and English reciprocal constructions would be rationalized if we analyze them in perspective of general collective morphemes ((105)) and their cross-linguistic variations (**Table 8**). This analysis allows us to capture the enigmatic contrast between Arabic and English described just below (104). The English symmetrical verb *kiss* in (101b') is accompanied by a phonetically empty collective morpheme $-\emptyset_{\text{COLL}}$ and licenses $[e]_{\text{RECIP}}$:

(101b') English: They_{<1+2>} *kissed*- $\emptyset_{\text{COLL}}$ $[e]_{\text{RECIP}<1\leftrightarrow 2>}$ (#in turn).

It therefore behaves on a par with the Arabic *t-marked kiss* in (100a'):

(100a') Arabic: They_{<1+2>} *t-kissed* $[e]_{\text{RECIP}<1\leftrightarrow 2>}$ (#in turn).

The “plain” *kiss* in (99a'), on the other hand, is disallowed in Arabic because of the lack of $\emptyset_{\text{COLL}}$ in this language, contrary to (101b') in English, which can involve $-\emptyset_{\text{COLL}}$ as in (106):

(99a') Arabic: *They_{<1+2>} *kissed* $[e]_{\text{RECIP}<1\leftrightarrow 2>}$.

Both (99b') and (102b') are disallowed but for distinct reasons.

(99b') Arabic: *They_{<1+2>} *helped* $[e]_{\text{RECIP}<1\leftrightarrow 2>}$.

(102b') English: They_{<1+2>} *helped* _____ *_{<1\leftrightarrow 2>}.

The “plain” *help* in (99b') is disallowed in Arabic again because of the lack of $\emptyset_{\text{COLL}}$, while it is disallowed in (102b') in English because $-\emptyset_{\text{COLL}}$ cannot be licensed by an *asymmetrical* verb, and hence $[e]_{\text{RECIP}}$ cannot be licensed in this language.

Once again, this has been just a quick and tentative sketch, but it may open up a possibility of cross-linguistic surveys on reciprocity which can be pursued on a much larger scale in the

necessarily advocate the presence of $[e]_{\text{RECIP}}$ as a complement.

- (i) a. $[\textit{Trump and Putin}]_1$ met with *Pope Francis*₂ $[_{\text{Ajet}}$ purely on behalf of { **himself**₂ / **themselves**₁ }].
- b. Trump₁ and Putin₂ met _____ $[_{\text{Ajet}}$ purely on behalf of **themselves**_{<1+2>}].

future.

Appendix C — T-morphemes in various Arabic dialects — An initial investigation:

This appendix reports the results of our cursory investigation of cross-dialectal variations in Arabic. We conducted three different types of grammaticality judgement tests with one each native speaker of Saudi, Kuwaiti, Egyptian, and Palestinian dialects, in addition to our JA informants. The investigations here were all prompted by the comments made by another anonymous reviewer on various interpretations of reciprocal sentences in languages/dialects other than JA. **Table 9** describes the informants we consulted with.

Table 9: Informants in the investigation of various Arabic dialects.

Speaker	Number	Gender	Age	Number of linguists involved
Jordanian	5	2 females • 3 males	20s - 1 30s - 3 50s - 1	1
Saudi • Palestinian • Kuwaiti • Egyptian	1 each, totaling 4	3 females • 1 male	30s - 4	2

Appendix C i — Distributivity with T-morphemes:

We first examined if distributive interpretations (i.e., non-overlapping subeventualities) are incompatible with T-marked verbs in other dialects just as in JA. We presented the sentences as in (108)–(110) below (in each respective dialect of Arabic) with two potentially compatible circumstances describing those sentences — one illustrates a collective situation (as in *a*) and the other illustrates a distributive situation (as in *b*). We then asked the speakers to pick the circumstances that are compatible with the provided sentence (and later to let us know if the unselected circumstances were just dispreferred or incompatible).

- (108) Bader₁ w-ʕumar₂ *t-saabagu* maʕ baʕð-hum l-baʕð<1↔2>.
 Bader.NOM and-Omar.NOM T-raced with some-them.GEN the-some.GEN
 ‘Bader and Omar raced with each other.’

Possible circumstances:

a. **Collective:**

t-saabagu b-wagt waahad.
 T-raced.3P.PL in-time one
 ‘They raced on a single occasion.’

b. **Distributive:**

Bader *rakað* masaafit kiloh b-ʕafar dagaajeg w-baʕdha
 Bader ran distance kilometer in-ten minutes and-after
 b-ʕwajj ʕumar *rakað* nafs el-masaafih b-rubuʕ saaʕah.
 in-little Omar ran same the-distance in-quarter hour
 ‘Bader ran one kilometer in ten minutes and after a little bit Omar ran the same distance in a quarter of an hour.’

- (109) et-taadzreen<1+2> *t-faaʕalau* maʕ baʕð-hum l-baʕð<1↔2>.
 the-businessmen.DL.NOM T-haggled with some-them.GEN the-some.GEN
 ‘The two businessmen haggled (negotiated prices) with each other.’

Possible circumstances:

a. **Collective:**

t-faaʕalau ʕala ʔasʕaar eθ-θalaadzeh wi-s-sajjaarah b-wagt waahad.
 T-haggled.3P.PL on prices the-fridge and-the-car in-time one
 ‘They haggled on the prices of the fridge and the car on a single occasion.’

b. **Distributive:**

el-ʔawwal *faaʕal* eθ-θaani ʕala siʕir el-mazharijjeh
 the-first haggled the-second on price the-vase
 b-suug el-dʒumʕah el-ʔusbuuʕ el-maaði bas eθ-θaani *faaʕal*
 in-market the-Friday the-week the-last but the-second haggled
 el-ʔawwal ʕala siʕir el-lawhah b-suug el-dʒumʕah haað el-ʔusbuuʕ.
 the-first on price the-painting in-market the-Friday this the-week
 ‘The first businessman haggled with the second businessman on the price of a vase in the last week’s Friday market, but the second one haggled with the first on the price of a painting in this week’s Friday market.’

- (110) el-ʔaxu₁ w-ʔuxt-uh₂ *t-ʕaawanu* maʕ baʕð-hum l-baʕð<1↔2>.
 the-brother.NOM and-sister-his.NOM T-helped with some-them.GEN the-some.GEN
 ‘The brother and his sister helped each other.’

Possible circumstances:

a. **Collective:**

t-ṣaawanu ṣala fuyl el-beet b-wagt waahad.
T-helped.3P.PL on work the-house in-time one
'They helped each other with the housework at the same time.'

b. **Distributive:**

el-ḡaxu *ḡaṣṭa* ḡuxt-uh maṣaari es-saneh el-maaḏjeh
the-brother gave sister-his money the-year the-last
w-hiih *wigfat* maṣaa-h lamma daxal el-mustaffa haaj es-saneh.
and-she supported with-him when entered the-hospital this the-year
'The brother gave his sister money last year and she supported him when he was hospitalized this year.'

All of the Jordanian, Saudi, Kuwaiti, Egyptian informants unanimously picked the collective circumstances and rejected the distributive circumstances in all of (108)–(110). That is, the reciprocal subeventualities involved in these sentences had to be interpreted as taking place only on a single occasion. The only variation we found was that the Palestinian speaker found (109) and (110) to be ambiguously interpretable while s/he agreed with the other speakers concerning (108) and found it to be strictly collective.

We also conducted the same test with (111) below, which an anonymous reviewer claims to permit distributivity (as described in *b*) in addition to collectivity (as described in *a*)

- (111) eljoom *t-ṣaanagna* ḡana w-ḡaxuu-j la-ḡawwal marrah.
today.ACC T-hugged I.NOM and-brother-my.NOM for-first.GEN time.GEN
'Today my brother and I hugged for the first time.'

Possible circumstances:

a. **Collective:**

ḡana₁ w-ḡaxuu-j₂ *t-ṣaanagna* maṣ baṣḏ-na l-baṣḏ_{<1↔2>}.
I and-brother-my T-hugged with some-us the-some
'My brother and I hugged each other.'

b. **(Allegedly) Distributive:**

ḡana w-ḡaxuu-j kul waahad fii-na *t-ṣaanag* maṣ ṣaaḥb-uh.
I and-brother-my each one of-us T-hugged with friend-his
'My brother and I each hugged respective friends.'

Although we did not think that the alleged “distributivity” as described in (111b) is equivalent to the distributivity we defined in terms of the lack of subeventuality-overlap, we had this sentence examined anyway together with (108)–(110). Then the speakers of all dialects rejected the (alleged) distributive interpretation as in (111b). Two of the informants even described this interpretation illogical (without our asking them to provide such a comment). **Table 10** summarizes all these results, whose clear trend alludes to the possibility that the T-morpheme requires collectivity not just in JA but in a fair number of dialects in Arabic.

Table 10: Summary of the investigation of distributive interpretations (various Arabic dialects).

	Jordanian	Saudi	Kuwaiti	Egyptian	Palestinian
(108)	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$
(109)	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	ok_a / ok_b
(110)	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	ok_a / ok_b
(111)	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$	$ok_a / \#b$

Appendix C ii — Distributivity in non-symmetrical reciprocals:

One may consider that “non-symmetrical reciprocals” as in (112) must always involve collective reciprocity and never permits distributivity.

- (112) a. The students followed *each other* into the room.
 b. The kids were chasing *each other* in the playground.

We believe, however, and our English-speaking informants agree, that non-symmetrical reciprocity can in fact hold involving *non-overlapping eventualities*. We must, though, manage to imagine some specific and fitting pragmatic contexts for such readings. The need for such extra endeavors perhaps turn distributive readings into non-default and possibly dispreferred interpretations in some cases. Some of such examples are:

- (113) a. The prisoners were so afraid of tipping-off and suspicious of other prisoners that they followed *each other* whenever someone headed toward the prison guards’ office.

- b. The top skiers are chasing *each other*'s records every time a competition is held.

We also examined how non-T-marked and T-marked verbs compare when they appear in non-symmetrical reciprocals in Jordanian, Saudi, Kuwaiti, and Palestinian dialects of Arabic although very few verbs seem to allow this construction in Arabic.³⁵ In all these dialects, the sentence involving a non-T-marked verb as in (114) below is compatible with either collective or distributive subeventualities as illustrated in *a* and *b*, respectively.

- (114) eṭ-ṭulaab₁₊₂ tabaʕu baʕḏ-hum l-baʕḏ_{<1↔2>}.
the-students.NOM followed some-them.ACC the-some.ACC
‘The students followed each other.’

a. **Collective:**

... lamma el-ḡustaaḏ ʕarax ʕalee-hum w-gallil-hum ʔiṭlaʕu ʕala eṣ-ṣaṭiḥ
... when the-teacher yelled at-them and-told-them go.up on the-roof
fawran la-tfillu min tsoonaami.
immediately to-escape from tsunami
‘... when the teacher yelled at them and told them to go up to the roof immediately to escape the tsunami.’

b. **Distributive:**

... lamma ṭaalib ʕimil blug el-xariif el-maaḏi w-ṭ-ṭulaab
... when student opened blog the-fall the-last and-the-students
eθ-θaanjiin ʕimlu bluggaat zaij-uh waḥad wara eθ-θaani.
the-other opened blogs like-him one after the-other
‘... when one student opened a blog account last fall and the others followed him by creating accounts one after the other.’

On the other hand, all speakers unanimously informed us that the T-marked verb in (115) below is compatible with the collective interpretation as in (114a) but not with the distributive interpretation as in (114b).

³⁵ Our informant of Egyptian Arabic told us that such a test cannot be conducted because this dialect uses two distinct morphologically unmarked verbs to express both meanings/sentences.

- (115) eṭ-ṭulaab₁₊₂ *t-taabaṣu* wara { baṣḍ-hum l-baṣḍ_{<1↔2>} / [e]_{RECIP<1↔2>} }.
the-students.NOM T-followed behind some-them.GEN the-some.GEN
‘The students followed each other.’

In short, the results of this test involving “non-symmetrical reciprocals” also alludes to the possibility that the T-morpheme requires collectivity not just in JA but in a fair number of other Arabic dialects.

Appendix C iii — Discontinuous reciprocity:

Finally, we also conducted a test to examine if various dialects of Arabic allow or disallow “discontinuous reciprocity” between the subject and the (comitative) complement. Just as we did in Section 2.2, we asked the informants whether or not a sentence negating reciprocity may follow such a “discontinuously reciprocal” sentence and make up a felicitous discourse, as in (116)–(119).

- (116) el-walad *t-ṣaawan* maṣ ʔuxt-uh, [w-*laakin* ʔuxt-uh
the-boy.NOM T-helped with sister-his.GEN, and-but sister-his.NOM
maa t-ṣaawanaṭ maṣ-uh].
NEG T-helped with-him.GEN

‘The boy helped his sister, [**but** his sister did **not** help him].’

- (117) el-walad *t-saamah* maṣ ʔuxt-uh, [w-*laakin* ʔuxt-uh
the-boy.NOM T-forgave with sister-his.GEN, and-but sister-his.NOM
maa t-saamahat maṣ-uh].
NEG T-forgave with-him.GEN

‘The boy forgave his sister, [**but** his sister did **not** forgive him].’

- (118) el-walad *t-saabag* maṣ ʔuxt-uh, [w-*laakin* ʔuxt-uh
the-boy.NOM T-raced with sister-his.GEN, and-but sister-his.NOM
maa t-saabagat maṣ-uh].
NEG T-raced with-him.GEN

‘The boy raced with his sister, [**but** his sister did **not** race with him].’

- (119) Bader *t-baaraz* maʕ Zeed, [w-laakin Zeed
 Bader.NOM T-fenced with Zeed.GEN, and-but Zeed.NOM
maa t-baaraz maʕ Bader].
 NEG T-fenced with Bader.GEN

‘Bader fenced with Zeed, [but Zeed did not fence with Bader].’

The results are summarized in **Table 11**, where the infelicity of the discourse (indicated by #) suggests the involvement of a “discontinuous reciprocal” interpretation in the first sentence while the felicity of the discourse (indicated by ok) suggests its absence.

Table 11: Summary of the investigation of discontinuous reciprocity (various Arabic dialects).

	Jordanian	Saudi	Kuwaiti	Egyptian	Palestinian	Verb	Verb type
(116)	ok	ok	ok	#	ok	<i>help</i>	Non-symmetrical
(117)	ok	ok	ok	#	ok	<i>forgive</i>	
(118)	#	ok	ok	#	#	<i>race</i>	Symmetrical
(119)	#	ok	ok	#	#	<i>fence</i>	

First, the speakers of all but Egyptian dialect rejected the “discontinuous reciprocal” interpretation of (116) and (117), which involve a non-symmetrical predicate. This may indicate that the T-morpheme does not induce reciprocity in any of these dialects except for the Egyptian dialect. Second, the speakers of all but Saudi and Kuwaiti dialects assigned the “discontinuous reciprocal” interpretation to (118) and (119), which involve a symmetrical predicate. In our post-hoc consultations, we obtained quite illuminating and intriguing comments from the Saudi and Kuwaiti speakers in regard to their judgments. They both informed us that they accepted (118) and (119), that is, they rejected the “discontinuous reciprocity” in these sentences based upon the interpretations such that *the brother* did but *the sister* did not intend to compete with the sibling in (118), and in (119), Bader fenced in his discretion but Zeed did not fence back, just defending himself from Bader’s thrusts. They also reported that (116) and (117)

are more clearly acceptable than (118) and (119). One possible interpretation of all the observations in **Table 11** then will be that in Jordanian, Saudi, Kuwaiti and Palestinian dialects, the reciprocal interpretation becomes available only when the involved verb inherently induces simultaneity independently of the use of the T-morpheme, although in the Egyptian dialect, the T-morpheme indeed seems to induce a reciprocal interpretation.

To sum up, the results of all three cross-dialectal investigations allude to the possibility that the function of the T-morpheme as a collectivizer rather than a reciprocalizer in JA can also be observed in a fair number of other Arabic dialects. Given that these investigations were conducted only in a small scale, however, we should not draw any firm conclusions at this point but await further investigations.

Appendix D — Interaction of collectivity and reciprocity:

Recall that, in Section 3, we proposed to capture the semantics of collectivity postulating the two operators *Coll* and *Dist* as in (120)–(121) below, which show up in a sentence as in (122).

$$(120) \quad \llbracket Dist \rrbracket^g = \lambda P_{\langle e, et \rangle} \lambda Z \lambda e. \forall x \sqsubseteq_A Z, P(x)(e) \quad (= (33b))$$

$$(121) \quad a. \quad \llbracket Coll_{Spatiotemporal} \rrbracket^g = \lambda P \lambda Z \lambda e [P(e)(Z) \ \& \ \forall e_1, e_2 \leq e \\ \llbracket [\exists y P(e_1)(y) \ \& \ \exists x P(e_2)(x)] \rightarrow K(e_1) \circ K(e_2) \rrbracket]$$

$$b. \quad \llbracket Coll_{Temporal} \rrbracket^g = \lambda P \lambda Z \lambda e [P(e)(Z) \ \& \ \forall e_1, e_2 \leq e \\ \llbracket [\exists y P(e_1)(y) \ \& \ \exists x P(e_2)(x)] \rightarrow \tau(e_1) \circ \tau(e_2) \rrbracket] \quad (= (32a-b))$$

$$(122) \quad [_{TP} \text{ Sbj}_1 [_{CollP} (Coll) [_{DistP} (Dist) [_{VP} t_1 V \text{ Complement }]]]] \quad (= (34))$$

In this appendix, we will attempt to figure out how the semantic interpretation is derived when collectivity is combined with (strong) reciprocity in a JA sentence. In doing so, we will appeal to some specific semantic analysis of reciprocity available in the literature which can be naturally couched in the semantic model for the collectivity we have adopted above, although we do not have any intention to make a strong commitment to it.

Presumably, when the overt distributive quantifier like *each* shows up in the sentence, *Dist*

must also appear so that they can be associated with each other in one way or another for proper interpretation, as illustrated in (123) for English and JA.

(123) a. John and Mary ***Dist each*** sang a song.

|---|

b. Zeed w-Bader ***Dist*** ***kul*** ***waahad*** yanna ?uynijeh.
 Zeed.NOM and-Bader.NOM DIST each.NOM one.GEN sang song.ACC
 |-----|
 ‘Zeed and Bader each sang a song.’

We assume that the same situation arises when a reciprocal anaphor appears as in (124a–b).

(124) a. John and Mary ***Dist*** love ***each other***.

|-----|

b. l-waladeen_{<1+2>} ***Dist*** ?aawanu ***baʕð-hum*** *l-baʕð*_{<1↔2>}.
 the-boys.DL.NOM DIST helped some-them.ACC the-some.ACC
 |-----|
 ‘The two boys helped each other.’

A reciprocal anaphor, however, not only demands a plural antecedent and $\llbracket Dist \rrbracket^g(\alpha)$ but also induces reciprocity (or disjointness) (Heim & Lasnik & May 1991). Buring (2005: 206) proposes to capture such dual properties of a reciprocal anaphor appealing to its semantic characterization in (125a) below, in which the letters *r* (= range) and *c* (= contrast) represent the two indices on *each other*.

- (125) a. $\llbracket each\ other_{r,c} \rrbracket^g = \text{the } x \sqsubseteq_A g(r) \text{ s.t. } x \neq g(c)$
 b. $\llbracket baʕð-hum\ l-baʕð_{r,c} \rrbracket^g = \text{the } x \sqsubseteq_A g(r) \text{ s.t. } x \neq g(c)$
 c. $\llbracket [e]_{r,c} \rrbracket^g = \text{the } x \sqsubseteq_A g(r) \text{ s.t. } x \neq g(c)$

When a reciprocal anaphor appears in sentences like (124a–b), it comes to denote, first, the individual sum of its antecedent (indicated by the range $g(r)$). Second, under the auspices of *Dist* in (122), *each/baʕð-hum* marks the presence of the atomic individuals of this range, which also identifies contrasts (indicated by $g(c)$). Finally, each of such contrasts and any atomic individual (*x*) within the same range are prescribed to play a role in the involved plural eventuality in a

disjoint (i.e., non-reflexive) manner ($x \neq g(c)$).³⁶ Once (125a) is adopted, it is not out of the question to consider that exactly the same analyses can be extended to the reciprocal anaphor *baʕð-hum l-baʕð* ‘each other’ in JA and even to the feature bundles involved in its covert counterpart as in (125b–c).

While we leave open whether and how exactly reciprocal anaphors should be decomposed and analyzed, we consider that *Dist* and an overt distributive element like *each* should be associated with each other in one way or another. Three feasible analyses to fulfil that goal come to mind. First, it can be assumed that *each* itself (rather than *Dist*) has the semantic denotation specified in (120) and it is covertly raised to *Dist* from its base-generated (and surface) position, in a way reminiscent of Heim & Lasnik & May’s (1991) LF raising of *each*, though the landing site assumed here is distinct. Second, *each* specified as in (125) is base-generated as the head *Dist* and lowered to its surface position in the course of the derivation to PF, by something akin to Affix Hopping as a morphological rule. Finally, *each* is base-generated at its surface position and undergoes in-situ association at LF with the *Dist* head as specified in (120) and located as in (122). It is reminiscent of Chomsky’s (2000) operation of Agree, though its motivation, involved mechanics and application timing are all different. Moreover, we hypothesize that *Coll* and T-morpheme in JA must also undergo similar association with each other, again in one of the following settings: (i) *t-* itself is specified as in (121) and undergoes covert raising to a higher empty position where *Coll* shows up in (122), (ii) *t-* as specified in (121) is the base-generated *Coll* head and lowered onto *Dist* and the verb via Affix Hopping, or (iii) *t-* is morphologically introduced on the verb in the lexicon and undergoes in-situ association at LF with the *Coll* head

³⁶ We are interpreting Buring’s analysis in the context of the event semantics. Buring assumes that *each* itself is *Dist* but we have postulated *Dist* as a functional head independent of *each*.

as specified in (121) and located as in (122). We leave it open at this point which (combination of) options would be optimal, though we advance our analysis provisionally with the semantic characterization of *Dist* in (120) and that of *Coll* as in (121).

Finally, appealing to and extending the semantic apparatuses adopted above, let us lay out a semantic analysis of “collective reciprocals” in JA as in (126) below, which can showcase all of the adopted apparatuses.

- (126) l-ʔaxu₁ w-ʔuxt-uh₂ t-ʕaawanu [e]_{RECIP <1↔2>}
the-brother.NOM and-sister-his.NOM T-helped
{ ^{ok}ʕala tarkiib s-sariir / # — huuh rakkabi-l-ha
on assembling.GEN the-bed.GE he.NOM assembled-for-her.GEN
s-sariir w-baʕdeen hiih naʔʔafat-l-uh l-yurfah }.
the-bed.ACC and-later she.NOM clean-for-him.GEN the-room.ACC
‘The brother and his sister helped each other { ^{ok}with assembling the bed / # — he assembled the bed for her and later she cleaned the room for him }.’

As the translation indicates, this seemingly intransitive sentence necessarily involves reciprocity, which we have ascribed to the presence of an empty reciprocal anaphor. At the same time, as the compatibility and incompatibility with the highlighted adjuncts indicate, the sentence involves obligatory collectivity induced by morphological T-marking. We believe that the plausibility of our overall analysis can be further illustrated when we can provide a proper semantic analysis of this “collective reciprocity” in JA combining our morpho-syntactic analyses with the semantic apparatuses as in (120)–(122) and (125). In (128) below, we provide the semantic computation of the “collective reciprocal” in (127). Here, we follow Buring (2005: 205) and postulate “binder prefixes β_1 and β_2 ” to provide the range ($g(r)$) and the contrast ($g(c)$), respectively, as in (127).

- (127) [TP l-ʔaxu₁ w-ʔuxt-uh₂ [CollP *Coll* [β_1 [DistP *Dist* [β_2 [VP t-ʕaawanu [e]_{<1↔2>}]]]]]].
the-brother.NOM and-sister-his.NOM T-helped
‘The brother and his sister helped each other.’

- (128) a. $\llbracket \text{T-helped} \rrbracket^g = \lambda x \lambda y \lambda e. \text{e is an event of y helping x}$
 b. $\llbracket \text{T-helped } [e]_{1,2} \rrbracket^g = \lambda y \lambda e. \text{e is an event of y helping the } x \sqsubseteq_A g(1) \text{ s.t. } x \neq g(2)$
 c. $\llbracket \beta_2 [\text{T-helped } [e]_{1,2}] \rrbracket^g = \lambda y \lambda e. \text{e is an event of y helping the } x \sqsubseteq_A g(1) \text{ s.t. } x \neq y$
 d. $\llbracket \text{Dist } [\beta_2 [\text{T-helped } [e]_{1,2}]] \rrbracket^g = \lambda Z \lambda e. \forall y \sqsubseteq_A Z, \text{e is an event of y helping the } x \sqsubseteq_A g(1) \text{ s.t. } x \neq y$
 e. $\llbracket \beta_1 [\text{Dist } [\beta_2 [\text{T-helped } [e]_{1,2}]]] \rrbracket^g = \lambda Z \lambda e. \forall y \sqsubseteq_A Z, \text{e is an event of y helping the } x \sqsubseteq_A Z \text{ s.t. } x \neq y$
 f. $\llbracket \text{Coll } [\beta_1 [\text{Dist } [\beta_2 [\text{T-helped } [e]_{<1 \leftrightarrow 2>}]]]] \rrbracket^g = \lambda Z \lambda e. \forall y \sqsubseteq_A Z, \text{e is an event of y helping the } x \sqsubseteq_A Z \text{ s.t. } x \neq y \wedge \forall e_1, e_2 \leq e$
 $\llbracket [\exists y P(e_1)(y) \wedge \exists x P(e_2)(x)] \rightarrow K(e_1) \circ K(e_2) \rrbracket$
 g. $\llbracket \text{TP} \rrbracket^g = \forall y \text{ s.t. } y \sqsubseteq_A \text{brother} \sqcup \text{sister}, \text{e is an event of y helping the } x \sqsubseteq_A \text{brother} \sqcup \text{sister s.t. } x \neq y \wedge \forall e_1, e_2 \leq e \llbracket [\exists y P(e_1)(y) \wedge \exists x P(e_2)(x)] \rightarrow K(e_1) \circ K(e_2) \rrbracket$

The predicate *t-helped* in (128a) maps some Agent individual (y) to some Theme individual (x) and there is some event (e) of the Agent helping the Theme. Next, the empty reciprocal anaphor *[e]* (each other) in (128b) enters the derivation, indicating that the atomic members of the plural subject as the two arguments involved in the event are distinct from each other. Then, β_2 introduced in (128c) identifies the contrast *g(2)* as *y*. The *Dist* operator in (128d) indicates that for each of the atomic members of the plural subject, there is an event of her/him helping the other. (*Z* indicates a plurality.) Then, β_1 introduced in (128e) identifies the range *g(1)* as *Z*. When the *Coll* operator combines with the *Dist* Phrase (*DistP*), it ensures that the two subevents *y helping x* and *x helping y* overlap in both time and space, as indicated by (128f). Finally, the whole sentence (TP) returns 1 if and only if the brother and his sister collectively helped each other, i.e., in shared time and space, as illustrated in (128g). Thus, we can properly compute the semantics of “collective reciprocity” in JA when we postulate *Dist* and *Coll* operators with their event-based semantic characterization, and combine them with our hypothesis that morphological T-marking introduces an empty reciprocal anaphor and obligatory collectivity into a seemingly intransitive construction.

Appendix E — Further examples of sentences and T-marked verbs:

- (71) c. **l-walad₁** *ħaka* [A_{jet} *ħan* *tadriib* { *šħaab-uh* / ***nafs-uh₁** } *karaati*].
the-boy.NOM talked about training.GEN friends-his.GEN self-him.GEN karate.ACC
‘The boy talked about the training of { his friends / *himself } in karate.’

- (72) c. **l-mudiireen₁** *gaabalu* **l-muawaḏḏafah₂**
the-managers.M.DL.NOM met the-employee.F.ACC
[A_{jet} *bi-xşuuş* *taḥsiin* { ***nafs-hum₁** / **nafs-ha₂** }].
in-regard.GEN betterment.GEN self-them.GEN self-her.GEN
‘The two male managers met with the female employee about the betterment of { *themselves / herself }.’

- (73) c. **l-mudiireen₁** *t-gaabalu* *maſ* **l-muawaḏḏafah₂**
the-managers.M.DL.NOM T-met with the-employee.F.ACC
[A_{jet} *bi-xşuuş* *taḥsiin* { ***nafs-hum₁** / **nafs-ha₂** }].
in-regard.GEN betterment.GEN self-them.GEN self-her.GEN
‘The two male managers met with the female employee about the betterment of { *themselves / herself }.’

- (74) c. **l-mudiireen<1+2>** *gaabalu* **baſḏ-hum** **l-baſḏ<1↔2>**
the-managers.M.DL.NOM met some-them.ACC the-some.ACC
[A_{jet} *bi-xşuuş* *taḥsiin* **nafs-hum<1+2>**].
in-regard.GEN betterment.GEN self-them.GEN
‘The two male managers met with each other about the betterment of themselves.’

- (75) c. **l-mudiireen<1+2>** *t-gaabalu* *maſ* **baſḏ-hum** **l-baſḏ<1↔2>**
the-managers.M.DL.NOM T-met with some-them.ACC the-some.ACC
[A_{jet} *bi-xşuuş* *taḥsiin* **nafs-hum<1+2>**].
in-regard.GEN betterment.GEN self-them.GEN
‘The two managers met with each other about the betterment of themselves.’

- (76) c. **l-mudiireen<1+2>** *t-gaabalu* [**e**]_{<1↔2>}
the-managers.M.DL.NOM T-met
[A_{jet} *bi-xşuuş* *taḥsiin* **nafs-hum<1+2>**].
in-regard.GEN betterment.GEN self-them.GEN
‘The two managers met with each other about the betterment of themselves.’

- (77) c. **l-ʔustaað₁** **t-naffar**
the-teacher.M.NOM T-distanced
[A_{jet} min taṭwiir { ṭulaab-uh / ***nafs-uh₁** }].
from improvement.GEN students-his.GEN **self-him**.GEN
‘The male teacher distanced himself from the improvement of { his students / *himself }.’
- (78) c. **l-ʔustaað₁** naffar **nafs-uh₁**
the-teacher.M.NOM distanced self-him.ACC
[A_{jet} min taṭwiir { ṭulaab-uh / ^{ok}**nafs-uh₁** }].
from improvement.GEN students-his.GEN self-him.GEN
‘The male teacher distanced himself from the improvement of { his students / ^{ok}himself }.’
- (19) b. **li-wlaad** **t-haadʒamu** ʕa-leeh
the-boys.NOM T-attacked on-him.GEN
{ ^{ok}**b-nafs** **l-wagt** / #**b-ʔawgaat** **muxtalfeh** }.
in-same.GEN the-time.GEN in-times.GEN different.GEN
‘The boys attacked him { ^{ok}at the same time / #at different times }.’

Table 2’: Further examples of T-morphology in JA.

	Input Verb Form	Derived Verb Form	Root	Derived Meanings
a.	2	5	<i>ḥ-m-m</i> ‘bathe’ <i>ʕ-l-m</i> ‘teach’ <i>ʔ-d-b</i> ‘behave’ <i>x-b-j</i> ‘hide’	<i>t-ḥammamu</i> ‘they bathed { themselves /#each other}’ <i>t-ʕallamu</i> ‘they taught { themselves /#each other}’ <i>t-ʔaddabu</i> ‘they behaved { themselves /#each other}’ <i>t-xabbu</i> ‘they hid { themselves /#each other}’
b.	3	6	<i>s-m-ḥ</i> ‘forgive’ <i>ʃ-dʒ-r</i> ‘fight’ <i>ʕ-t-b</i> ‘blame’ <i>ʕ-n-g</i> ‘hug’	<i>t-saamaḥu</i> ‘they forgave { each other /#themselves}’ <i>t-ʃaadʒaru</i> ‘they fought { each other /#themselves}’ <i>t-ʕaatabu</i> ‘they blamed { each other /#themselves}’ <i>t-ʕaanagu</i> ‘they hugged { each other /#themselves}’
c.	1	8	<i>r-m-j</i> ‘throw’ <i>ʕ-d-l</i> ‘edify’ <i>r-g-j</i> ‘elevate’ <i>r-dʒ-l</i> ‘act manly’	<i>ʔir-t-amu</i> ‘they threw { themselves /#each other}’ <i>ʔiʕ-t-adalu</i> ‘they edified { themselves /#each other}’ <i>ʔir-t-agu</i> ‘they elevated { themselves /#each other}’ <i>ʔir-t-aɖʒalu</i> ‘they acted manly { themselves /#each other}’
d.	1	8	<i>f-r-g</i> ‘separate’ <i>ɖʒ-m-ʕ</i> ‘meet’ <i>x-l-f</i> ‘disagree’ <i>x-l-t</i> ‘mix’	<i>ʔif-t-ragu</i> ‘they separated from { each other /#themselves}’ <i>ʔiɖʒ-t-amaʕu</i> ‘they met { each other /#themselves}’ <i>ʔix-t-alafu</i> ‘they disagreed { each other /#themselves}’ <i>ʔix-t-alatu</i> ‘they mixed { each other /#themselves}’

- (43) b. l-ṭixwah t-xabbu bi-s-siddeh.
the-brothers.NOM T-caused.to.hide in-the.attic.GEN

‘The brothers caused themselves to hide into the attic. (= The brothers hid themselves into the attic.)’

(46') F3-transitive:

- a. l-walad qaasam *(šaahb-uh) ʕa-l-keek.
the-boy.NOM split friend-his.ACC on-the-cake.GEN

‘The boy split (the shares of) the cake with his friend. (= The boy and his friend split the cake.)’

- b. l-walad saabag *(šaahb-uh) l-bint.
the-boy.NOM raced friend-his.ACC the-girl.ACC

Intended: ‘The boy raced the girl with his friend. (= Both the boy and his friend raced the girl.)’

- (51) c. l-bint t-ḏaahakat *(maʕ l-walad).
the-girl.NOM T-laughed with the-boy.GEN

‘The girl laughed with the boy. (= Both the girl and the boy laughed.)’

- (55) c. l-binteen t-ḏaahaken { ^{ok}sawijjeh / #wahadeh baʕd θ-θaanjeh }.
the-girls.DL.NOM T-laughed together.ACC one.ACC after the-second.GEN

‘Lit: The two girls laughed { ^{ok}together / #one after another }.’

- (55') c. l-binteen_{<1+2>} t-ḏaahaken [e]_{<1↔2>} { ^{ok}sawijjeh / #wahadeh baʕd θ-θaanjeh }.
the-girls.DL.NOM T-laughed together.ACC one.ACC after the-second.GEN

‘Each of the two girls laughed with the other { ^{ok}together / #one after another }.’

- (56) c. l-binteen ḏaahaken *(l-walad).
the-girls.DL.NOM laughed the-boy.ACC

‘The two girls laughed with the boy.’

- (57) c. Bader t-baaraz maʕ Bandar { ^{ok}b-nafs l-wagt /
Bader T-fenced with Babdar in-same.GEN the-time.GEN
#bi-t-tanaawub }.
in-the-turn.GEN

‘Bader fenced with Bandar { ^{ok}at the same time / #in turn }.’

- (63) c. l-dḡunuud ʔin-t-aharu { ^{ok}b-nafs el-wagt / ^{ok}bi-t-tanaawub }.
the-soldiers.NOM T-suicided in-same.GEN the-time.GEN in-the-turn.GEN

‘The soldiers suicided { ^{ok}simultaneously / ^{ok}in turn }.’

- (64) c. l-waladeen<1+2> ʔidɟ-t-amaʃu [e]<1↔2> bi-ʃ-ʃaariʃ { ^{ok}b-nafs l-wagt /
the-boys.DL.NOM T-gathered in-the-street.GEN in-same.GEN time.GEN
#waahad baʃd θ-θaani / #bi-t-tanaawub }.
one.ACC after the-second.GEN in-the-turn.GEN
‘Each of the two boys gathered with the other on the street { ^{ok}at the same time /
#one after the other / #in turn }.’
- (65) c. l-walad ʔidɟ-t-amaʃ *(maʃ l-bint)
the-boy.NOM T-gathered with the-girl.GEN
{ ^{ok}eð-ðuhur / #bi-t-tanaawub }.
the-noon.ACC in-the-turn.GEN
‘The boy gathered with the girl { ^{ok}at noon / #in turn }.’

References:

- Antonenko, Andrei. 2012. *Feature-based binding and phase theory*. Stony Brook, NY: Stony Brook University dissertation.
- Barker, Chris. 2012. Quantificational binding does not require c-command. *Linguistic Inquiry* 43(4). 614–633.
- Carroll, Susanne. 1986. On non-aphor reflexives. *Revue québécoise de linguistique* 15(2). 135-165.
- Chomsky, Noam 1981. *Lectures on Government and Binding*. Dordrecht: Foris Publications.
- Chomsky, Noam. 1986. *Knowledge of language: Its nature, origin, and use*. New York: Praeger.
- Chomsky, Noam. 2000. Minimalist inquiries: The framework. In Martin, Roger & Michaels, David & Uriagereka, Juan (eds.), *Step-by-step: Essays on minimalist syntax in honor of Howard Lasnik*, 89–156. Cambridge, MA: MIT Press.
- Dench, Alan Charles. 1995. *Martuthunira: A language of the Pilbara Region of Western Australia* (Pacific Linguistics Series C-125). Canberra: The Australian National University.
- Dixon, Robert M. W. 1988. *A grammar of Boumaa Fijian*. Chicago: The University of Chicago Press.
- Higginbotham, James. 1980. Pronouns and bound variables. *Linguistic Inquiry* 11. 679–708.
- Hornstein, Norbert. 2001. *Move! A minimalist theory of construal*. Malden, MA: Blackwell.
- Kayne, Richard S. 1994. *The antisymmetry of syntax*. Cambridge, MA: MIT Press.

Pollard, Carl & Sag, Ivan. 1992. Anaphors in English and the scope of the binding theory. *Linguistic Inquiry* 23. 261–305.

Varaschin, Giuseppe. 2020. Anti-reflexivity and logophoricity: an account of unexpected reflexivization contrasts. *Glossa* 5(1). 1–32.

References

- Bar-Asher Siegal, Elitzur A. 2016. The Semitic templates from the perspective of reciprocal predicates. *Proceedings of the 10th Mediterranean Morphology Meeting*, 16–30.
- Bar-Asher Siegal, Elitzur A. 2020. *The NP-strategy for Expressing Reciprocity: Typology, History, Syntax and Semantics*. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/tsl.127>
- Benmamoun, Elabbas. 2003. Reciprocals as plurals in Arabic. In Lecarme, Jacqueline (ed.), *Research in Afroasiatic grammar II: Selected Papers from the Fifth Conference on Afroasiatic Languages*, 53–62. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/cilt.241.05ben>
- Büring, Daniel. 2005. *Binding theory*. Cambridge: Cambridge University Press. DOI: <https://doi.org/10.1017/CBO9780511802669>
- Carlson, Greg. 1998. Thematic roles and the individuation of events. In Rothstein, Susan (ed.), *Events and grammar (studies in linguistics and philosophy 70)*, 35–51. Dordrecht: Kluwer Academic Publishers. DOI: https://doi.org/10.1007/978-94-011-3969-4_3
- Dalrymple, Mary & Kanazawa, Makoto & Kim, Yookyung & McHombo, Sam & Peters, Stanley. 1998. Reciprocal expressions and the concept of reciprocity. *Linguistics and philosophy* 21(2). 159–210. DOI: <https://doi.org/10.1023/A:1005330227480>
- Dimitriadis, Alexis. 2008a. The event structure of irreducibly symmetric reciprocals. In Dölling, Johannes & Heyde-Zybatow, Tatjana & Schäfer, Martin (eds.), *Event Structures in Linguistic Form and Interpretation*, 328–354. Berlin: Mouton de Gruyter. DOI: <https://doi.org/10.1515/9783110925449.327>
- Dimitriadis, Alexis. 2008b. Irreducible symmetry in reciprocal constructions. In König, Ekkehard & Gast, Volker (eds.), *Reciprocals and reflexives: Theoretical and typological explorations*, 375–409. Berlin: Mouton de Gruyter.
- Doron, Edit. 2003. Agency and voice: The semantics of the Semitic templates. *Natural Language Semantics* 11. 1–67. DOI: <https://doi.org/10.1023/A:1023021423453>
- Essien, Hannah. 2014. Structural study of Arabic morphology. Bloomington, IN: Indiana University Bloomington dissertation.
- Fassi Fehri, Abdelkader. 2012. *Key features and parameters in Arabic grammar*. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/la.182>
- Fischer, Wolfdietrich. 2002. *A grammar of Classical Arabic (third revised edition)*. New Haven, Connecticut: Yale University Press.
- Grodzinsky, Yosef & Reinhart, Tanya. 1993. The innateness of binding and coreference. *Linguistic Inquiry* 24. 69–102.
- Heim, Irene. 1998. Anaphora and semantic interpretation: A reinterpretation of Reinhart's approach. In Sauerland, Uli & Percus, Orin (eds.), *The interpretive tract: Working Papers in Syntax and Semantics, MIT Working Papers in Linguistics, Volume 25*, 205–246. Cambridge, MA, Department of Linguistics, MIT.
- Heim, Irene & Lasnik, Howard & May, Robert. 1991. Reciprocity and plurality. *Linguistic Theory* 22(1). 63–101.

- Higginbotham, James. 1985. On semantics. *Linguistic Inquiry* 16. 547–593.
- Holes, Clive. 2004. *Modern Arabic structures, functions, and varieties*. Washington, DC: Georgetown University Press.
- Laks, Lior. 2007. Morphology and thematic arity operations: Evidence from Standard Arabic. In Mughazy, Mustafa A. (ed.), *Perspectives on Arabic linguistics XX: Papers from the twentieth annual symposium on Arabic linguistics*, 51–67. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/cilt.290.06lak>
- Landman, Fred. 1989. Groups, I. *Linguistics and Philosophy* 12(5). 559–605. DOI: <https://doi.org/10.1007/BF00627774>
- Landman, Fred. 2000. *Events and plurality: The Jerusalem lectures (studies in linguistics and philosophy)*. Dordrecht: Kluwer Academic Publishers. DOI: <https://doi.org/10.1007/978-94-011-4359-2>
- Laserson, Peter. 1995. *Plurality, conjunction, and events (studies in linguistics and philosophy 55)*. Dordrecht: Kluwer Academic Publishers. DOI: https://doi.org/10.1007/978-94-015-8581-1_5
- Laserson, Peter. 1998. Events in the semantics of collectivizing adverbials. In Rothstein, Susan (ed.), *Events and grammar (studies in linguistics and philosophy 70)*, 273–292. Dordrecht: Kluwer Academic Publishers. DOI: https://doi.org/10.1007/978-94-011-3969-4_11
- LeTourneau, Mark S. 1998. Implicit reciprocals in Standard Arabic. In Benmamoun, Elabbas & Eid, Mushira & Haeri, Niloofar (eds.), *Perspectives on Arabic linguistics XI: Papers from the eleventh annual symposium on Arabic linguistics*, 91–113. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/cilt.167.07let>
- Link, Godehard. 1998. *Algebraic semantics in language and philosophy* (CSLI Lecture Notes No. 74). Stanford, California: CSLI Publications.
- Lønning, Jan Tore. 2011. Plurals and collectives. In van Benthem, Johan & ter Meulen, Alice (eds.), *Handbook of logic and language*, Second Edition, 989–1033. Amsterdam: Elsevier. DOI: <https://doi.org/10.1016/B978-0-444-53726-3.00023-2>
- Mari, Alda. 2014. Each other, asymmetry and reasonable futures. *Journal of Semantics* 31. 209–261. DOI: <https://doi.org/10.1093/jos/fft003>
- McCarthy, John Joseph. 1981. A prosodic theory of nonconcatenative morphology. *Linguistic Inquiry* 12(3). 373–418.
- McCarthy, John Joseph. 1982. *Formal problems in Semitic phonology and morphology*. Texas: University of Texas, Austin.
- Peres, Joao Andrade. 1998. Issues on Distributive and Collective Readings. In Hamm, Fritz & Hinrichs, Erhard (eds.), *Plurality and Quantification (Studies in Linguistics and Philosophy 69)*, 339–365. Dordrecht: Kluwer Academic Publishers. DOI: https://doi.org/10.1007/978-94-017-2706-8_9
- Reinhart, Tanya. 1983. *Anaphora and semantic interpretation*. London: Croom Helm.
- Reinhart, Tanya & Siloni, Tal. 2005. The lexicon-syntax parameter: Reflexivization and other arity operations. *Linguistic Inquiry* 36(3). 389–436. DOI: <https://doi.org/10.1162/0024389054396881>

- Ryding, Karin C. 2005. *A reference grammar of Modern Standard Arabic*. Cambridge: Cambridge University Press. DOI: <https://doi.org/10.1017/CBO9780511486975>
- Ryding, Karin C. 2014. *Arabic: A linguistic introduction*. Cambridge: Cambridge University Press. DOI: <https://doi.org/10.1017/CBO9781139151016>
- Sabato, Sivan & Winter, Yoad. 2012. Relational domains and the interpretation of reciprocals. *Linguistics and Philosophy* 35(3). 191–241. DOI: <https://doi.org/10.1007/s10988-012-9117-x>
- Siloni, Tal. 2001. Reciprocal verbs. In Falk, Yehuda N. (ed.), *Proceedings of the Israel association of theoretical linguistics* 17. Jerusalem: Hebrew University of Jerusalem. <http://linguistics.huji.ac.il/IATL/17/TOC.html>.
- Siloni, Tal. 2002. Active lexicon. *Theoretical Linguistics* 28. 383–400. DOI: <https://doi.org/10.1515/thli.28.3.383>
- Siloni, Tal. 2008. The syntax of reciprocal verbs: An overview. In König, Ekkehard & Gast, Volker (eds.), *Reciprocals and reflexives: Theoretical and typological explorations*, 451–498. Berlin: Mouton de Gruyter.
- Siloni, Tal. 2012. Reciprocal verbs and symmetry. *Natural Language & Linguistic Theory* 30(1). 261–320. DOI: <https://doi.org/10.1007/s11049-011-9144-2>
- Thackston, Wheeler M. 2000. *An introduction to Koranic and Classical Arabic: An elementary grammar of the Language*. Bethesda, Maryland: IBEX Publishers.
- Watson, Janet C. E. 2002. *The phonology and morphology of Arabic*. Oxford: Oxford University Press.
- Wright, William. 1896. *A grammar of the Arabic language*, Vol.1. Originally written in German by Carl Caspari and Translated by William Wright, Smith, William Robertson & de Goeje, Michael Jan (eds.). Cambridge: Cambridge University Press. (Original work published 1874.)

